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- (54) **MERGE CANDIDATE CONSTRUCTION**
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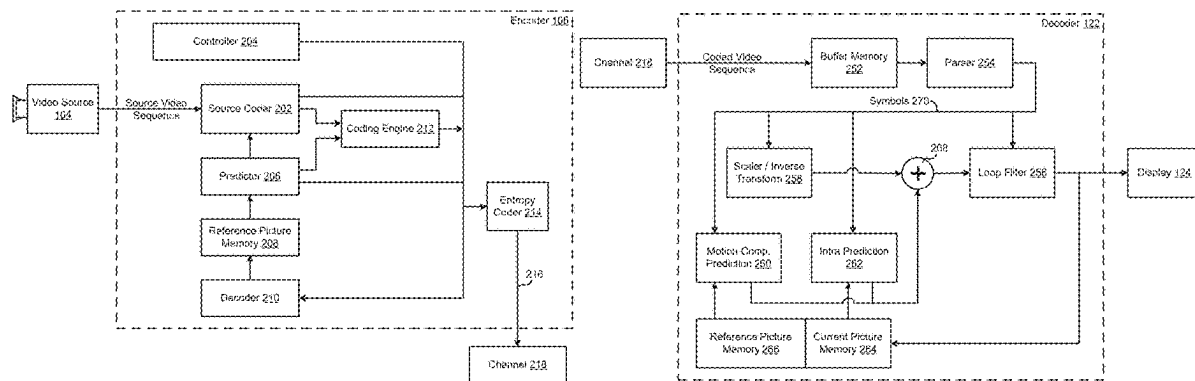
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(57) **ABSTRACT**

The various implementations described herein include methods and systems for coding video. In one aspect, a method includes receiving a video bitstream comprising a current picture composed of a plurality of blocks, including a current block. The method also includes identifying a merge candidate for the current block and identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique. The method further includes identifying a reference vector for the refined merge candidate and populating a merge candidate list for the current block using the reference vector. The method also includes reconstructing the current block using information from the merge candidate list.



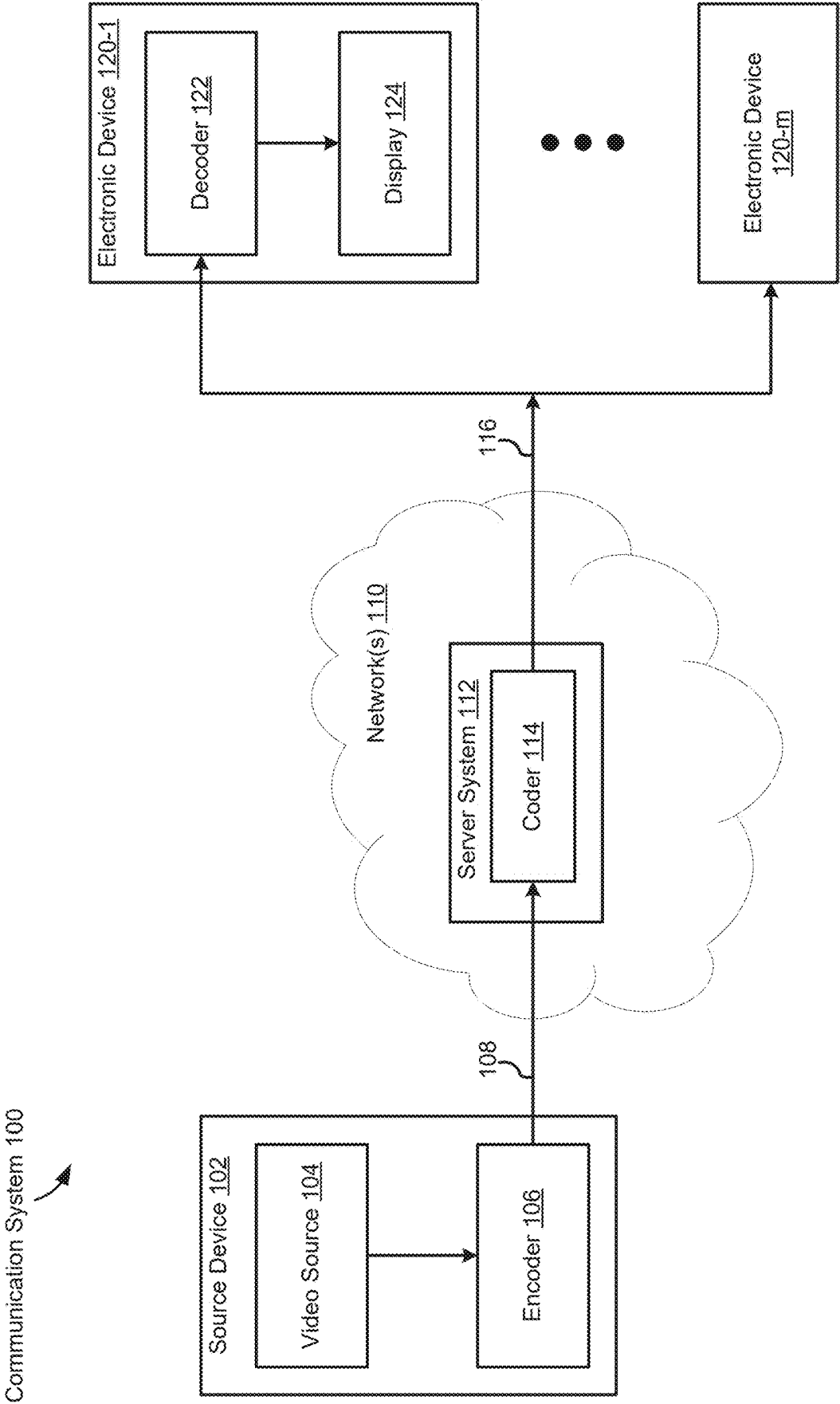


FIG. 1

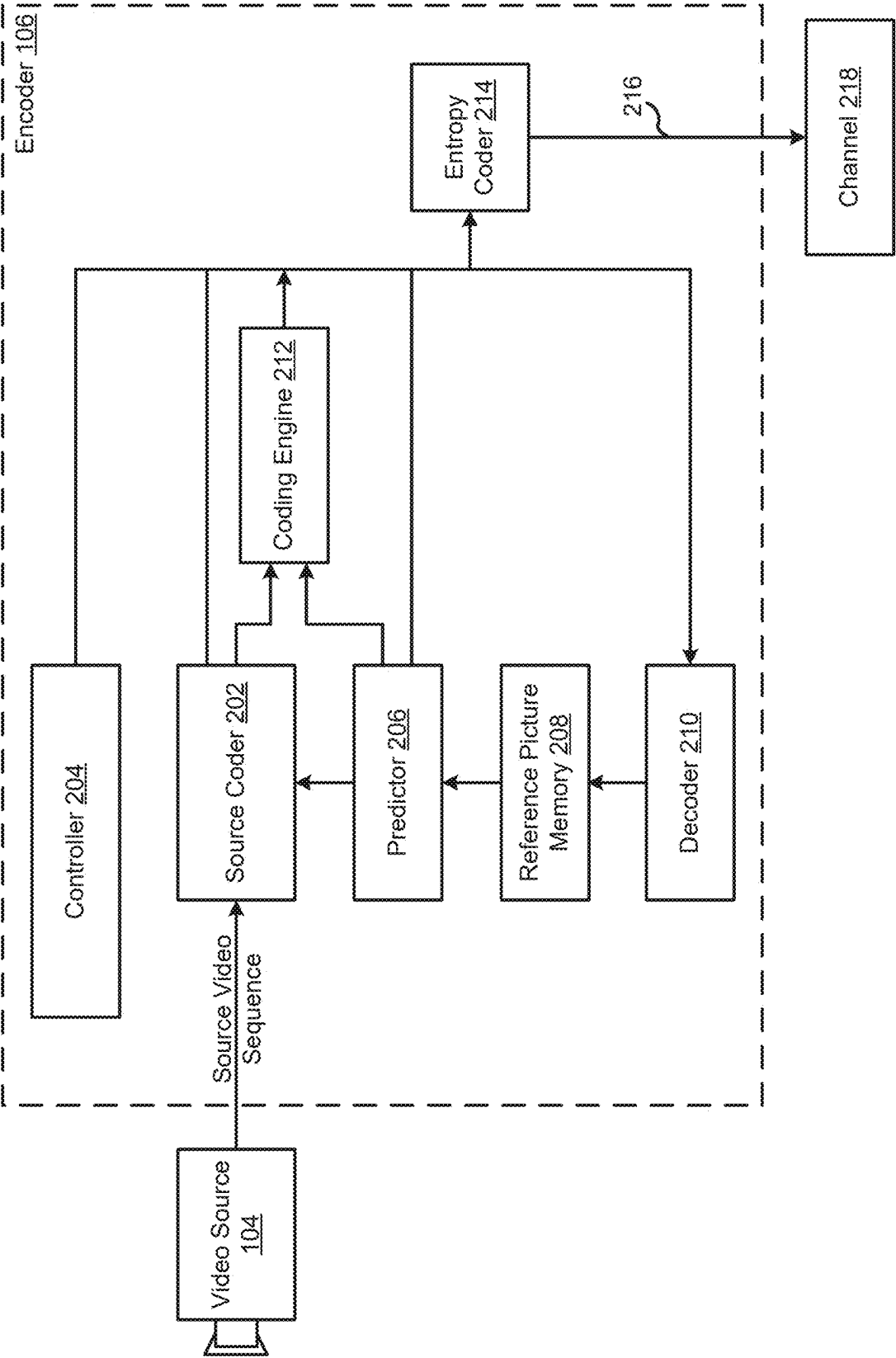


FIG. 2A

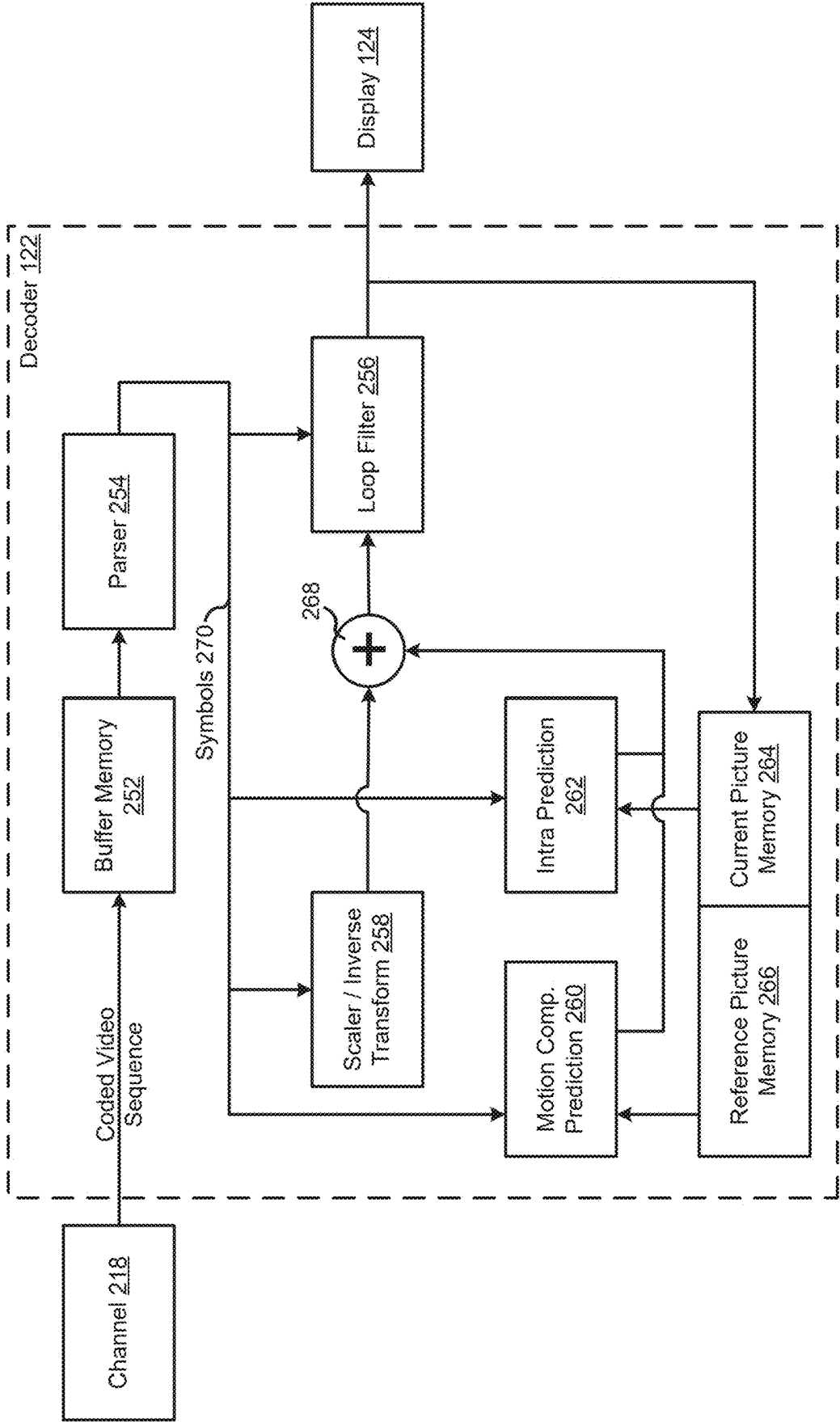


FIG. 2B

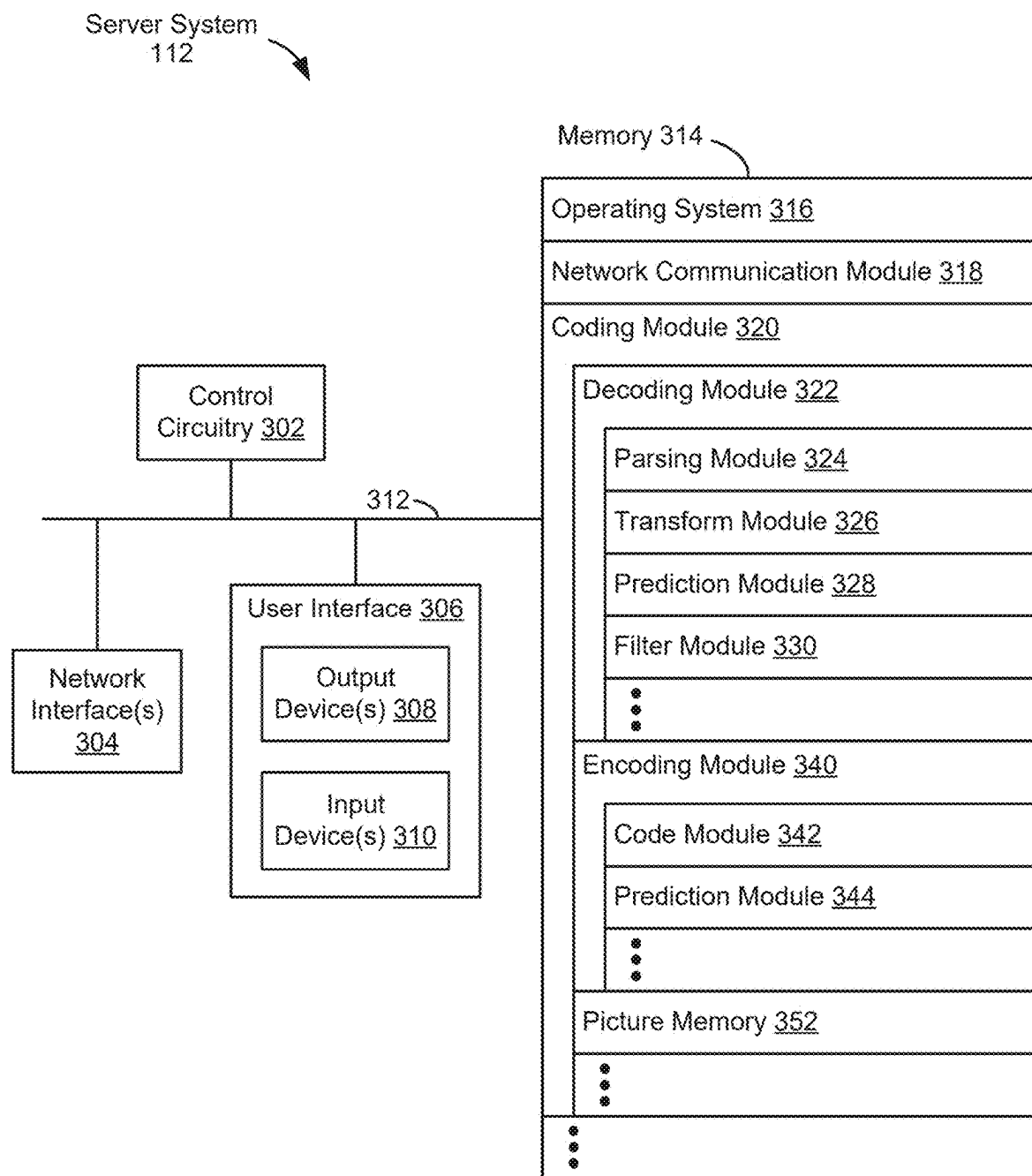


FIG. 3

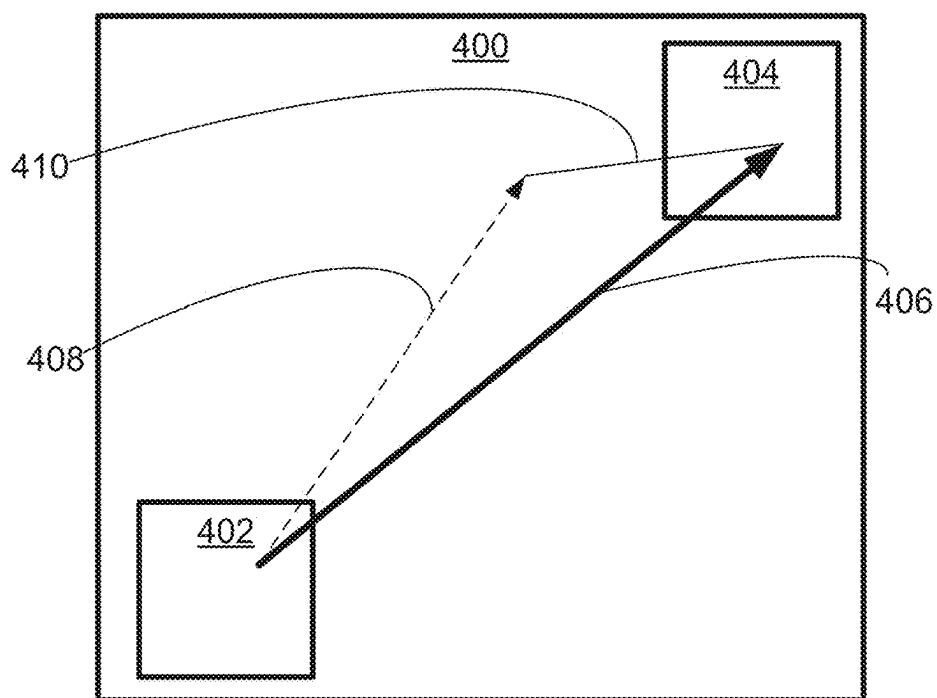


FIG. 4A

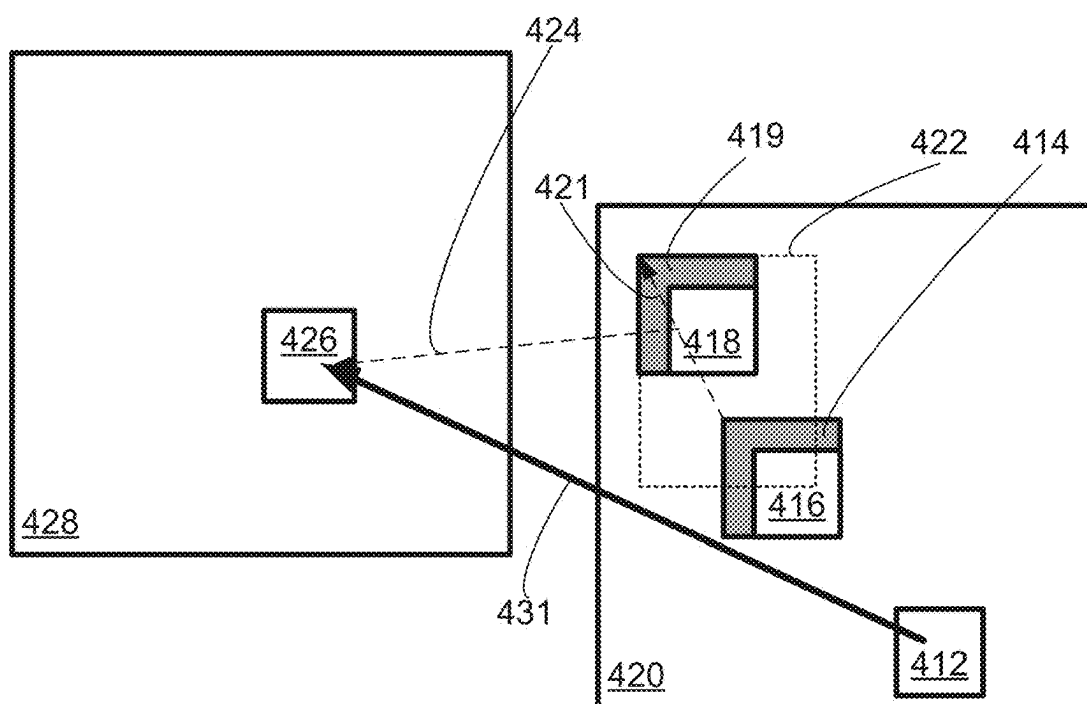
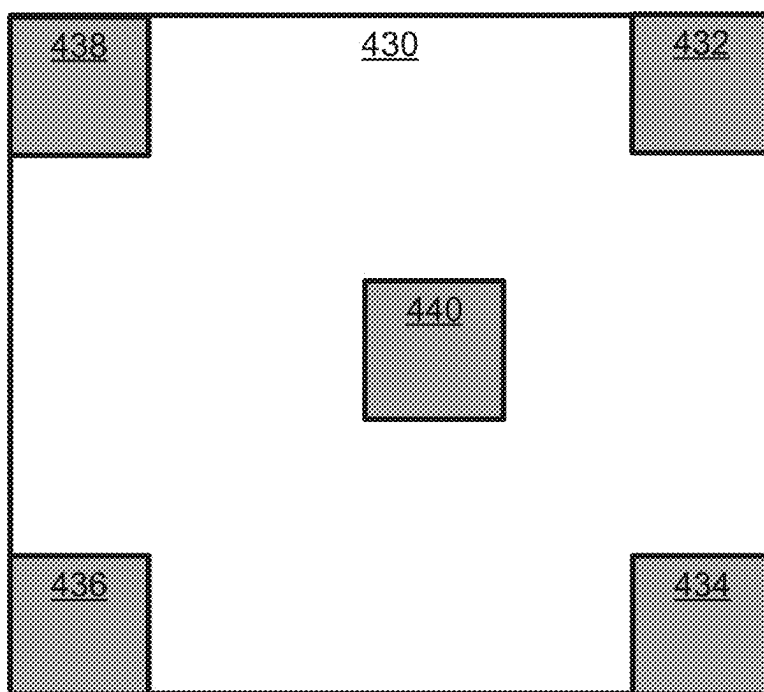
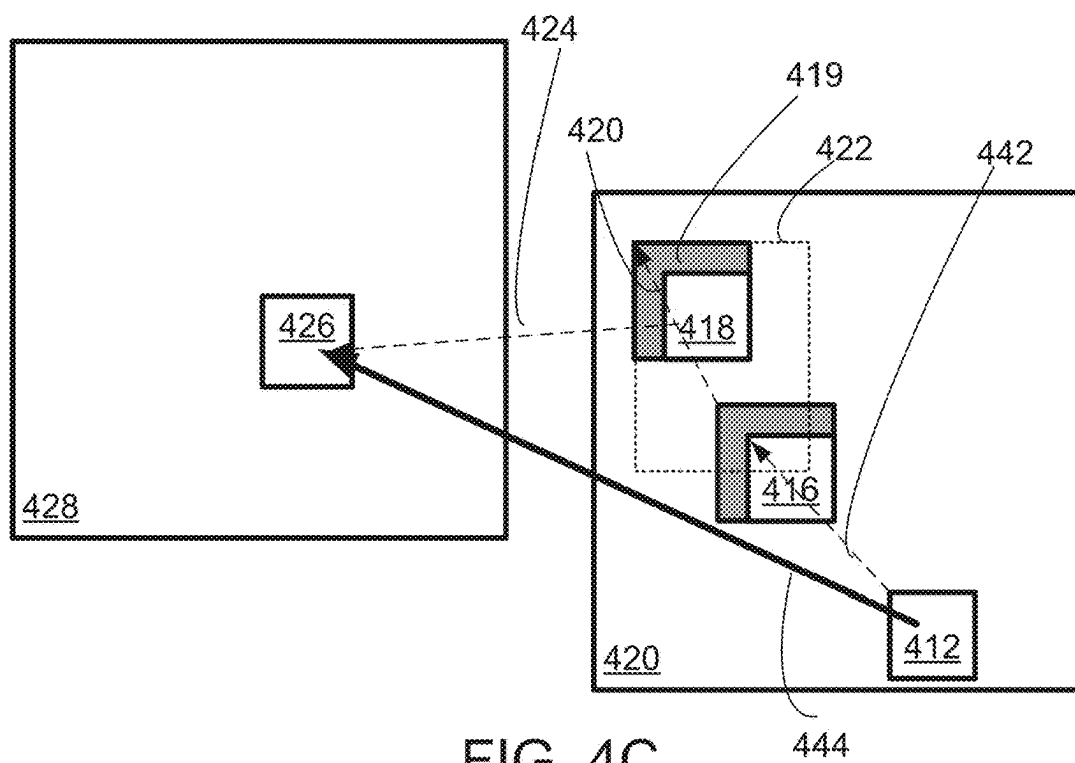


FIG. 4B



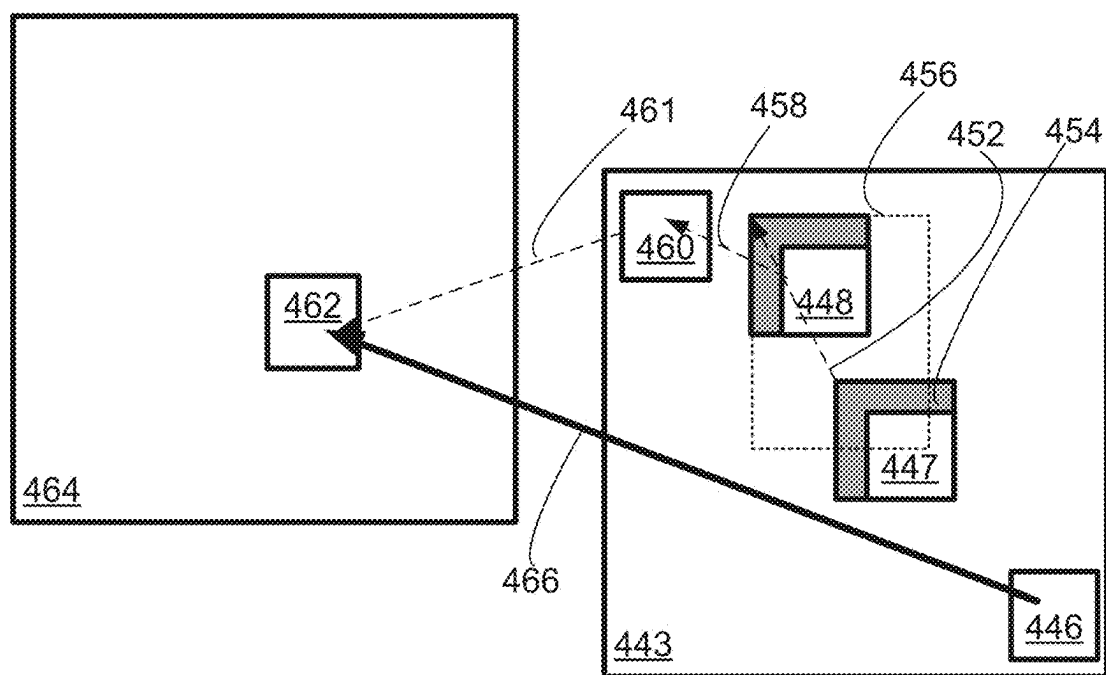


FIG. 4E

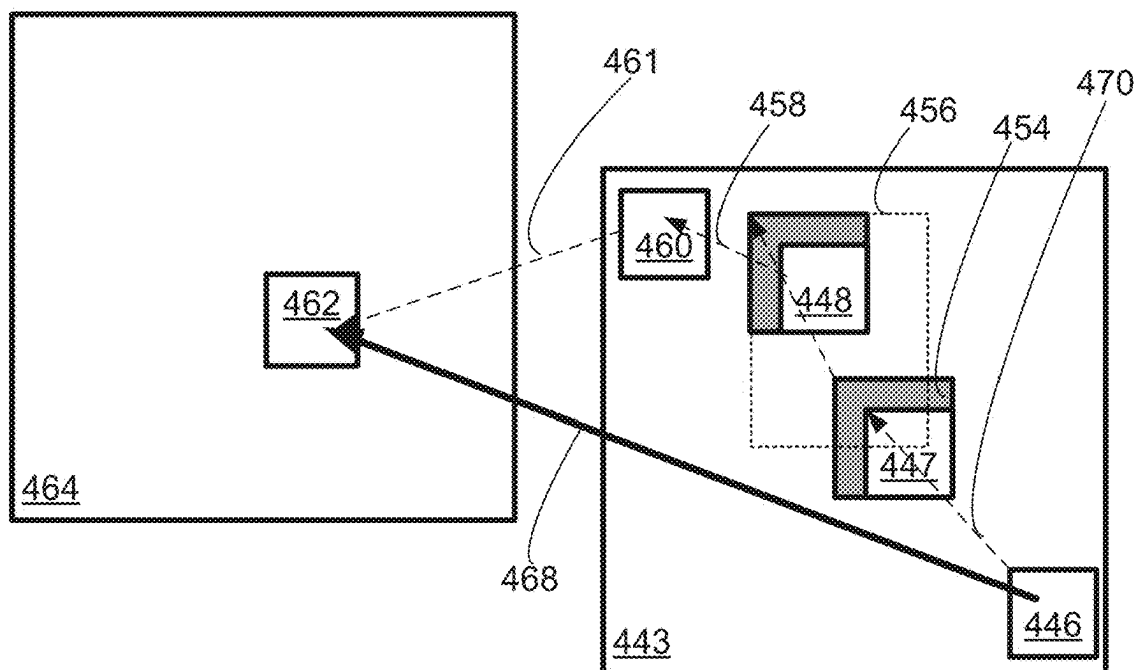


FIG. 4F

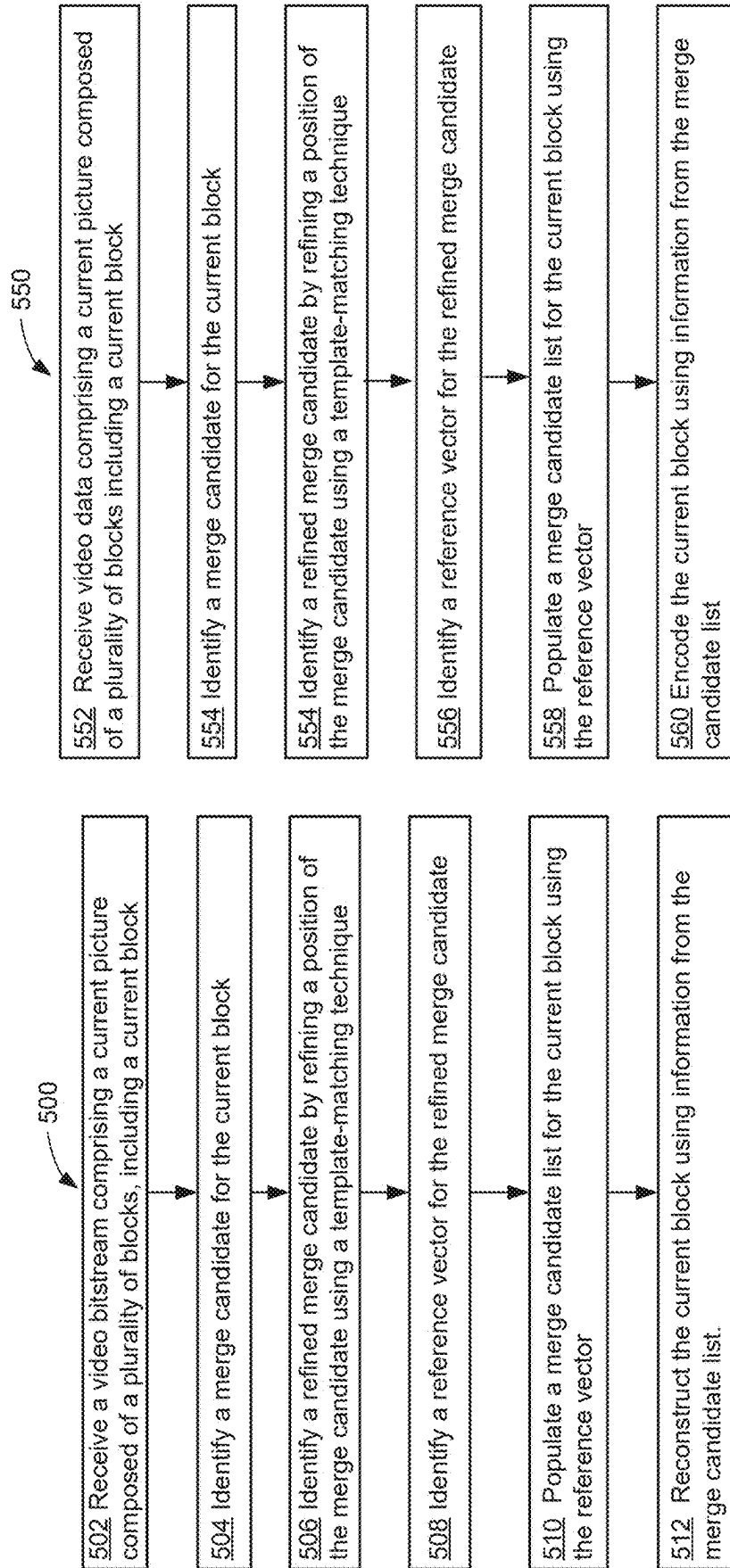


FIG. 5A

FIG. 5B

MERGE CANDIDATE CONSTRUCTION

PRIORITY

[0001] This application claims priority to U.S. Provisional Patent Application No. 63/562,059, entitled “Merge Candidate Construction” filed Mar. 6, 2024, which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] The disclosed embodiments relate generally to video coding, including but not limited to systems and methods for using template-matching to identify motion vector candidates.

BACKGROUND

[0003] Digital video is supported by a variety of electronic devices, such as digital televisions, laptop or desktop computers, tablet computers, digital cameras, digital recording devices, digital media players, video gaming consoles, smart phones, video conferencing devices, video streaming devices, etc. The electronic devices transmit and receive or otherwise communicate digital video data across a communication network, and/or store the digital video data on a storage device. Due to a limited bandwidth capacity of the communication network and limited memory resources of the storage device, video coding may be used to compress the video data according to one or more video coding standards before it is communicated or stored. The video coding can be performed by hardware and/or software on an electronic/client device or a server providing a cloud service.

[0004] Video coding generally utilizes prediction methods (e.g., inter-prediction, intra-prediction, or the like) that take advantage of redundancy inherent in the video data. Video coding aims to compress video data into a form that uses a lower bit rate, while avoiding or minimizing degradations to video quality. Multiple video codec standards have been developed. For example, High-Efficiency Video Coding (HEVC/H.265) is a video compression standard designed as part of the MPEG-H project. ITU-T and ISO/IEC published the HEVC/H.265 standard in 2013 (version 1), 2014 (version 2), 2015 (version 3), and 2016 (version 4). Versatile Video Coding (VVC/H.266) is a video compression standard intended as a successor to HEVC. ITU-T and ISO/IEC published the VVC/H.266 standard in 2020 (version 1) and 2022 (version 2). AOMedia Video 1 (AV1) is an open video coding format designed as an alternative to HEVC. On Jan. 8, 2019, a validated version 1.0.0 with Errata 1 of the specification was released. Enhanced Compression Model (ECM) is a video coding standard that is currently under development. ECM aims to significantly improve compression efficiency beyond existing standards like HEVC/H.265 and VVC, essentially allowing for higher quality video at lower bitrates.

SUMMARY

[0005] The present disclosure describes, amongst other things, a set of methods for video (image) compression, such as using intra template matching to populate a merge candidate list with one or more refined merge candidates, where the merge candidate list is used to reconstruct a current block. For example, a template-matching process may be applied to the current reconstructed picture to find adjacent and/or non-adjacent block(s) that have motion vector (MV)

and/or block vector (BV) information that can be further used by the current block. The MV, BV and or displacement vector (DV) from the adjacent or non-adjacent block may be used to construct a refined merge candidate list.

[0006] The techniques in the preceding paragraph can be used to improve coding accuracy, such as by increasing merge list diversity and using MV information from a prediction block that more closely matches the current block. Additionally, signaling overhead may be reduced by using the merge candidate list and signaling an index to the list as opposed to directly signaling a MV. Compression efficiency may also be improved by populating the merge candidate list more efficiently using the methods described herein.

[0007] In accordance with some embodiments, a method of video decoding is provided. The method includes (i) receiving a video bitstream (e.g., a coded video sequence) comprising a current picture composed of a plurality of blocks, including a current block and (ii) identifying a merge candidate for the current block; (iii) identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique. The method also includes (iv) identifying a reference vector for the refined merge candidate, (v) populating a merge candidate list for the current block using the reference vector; and (vi) reconstructing the current block using information from the merge candidate list.

[0008] In accordance with some embodiments, a method of video encoding includes (i) receiving video data (e.g., a source video sequence) comprising a current picture composed of a plurality of blocks, including a current block. The method includes (ii) identifying a merge candidate for the current block; (iii) identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique; (iv) identifying a reference vector for the refined merge candidate; (v) populating a merge candidate list for the current block using the reference vector; and (vi) encoding the current block using information from the merge candidate list.

[0009] In accordance with some embodiments, a method of video decoding includes (i) receiving a video bitstream comprising a current picture composed of a plurality of blocks, including a current block; (ii) identifying a merge candidate for the current block; (iii) identifying a reference vector for the merge candidate; (iv) generating a combined reference vector by combining the reference vector with a displacement vector; (v) populating a merge candidate list for the current block using the combined reference vector; and (vi) reconstructing the current block using information from the merge candidate list.

[0010] In accordance with some embodiments, method of processing visual media data includes (i) obtaining a source video sequence that comprises a plurality of frames; and (ii) performing a conversion between the source video sequence and a video bitstream of visual media data according to a format rule, where the video bitstream comprises a plurality of encoded blocks including a current block. The format rule specifies that: (a) a merge candidate is to be identified for the current block; (b) a refined merge candidate is to be identified by refining a position of the merge candidate using a template-matching technique; (c) a reference vector is to be identified for the refined merge candidate; (d) a merge candidate list is to be populated for the current block using

the reference vector; and (e) the current block is to be reconstructed using information from the merge candidate list.

[0011] In accordance with some embodiments, a computing system is provided, such as a streaming system, a server system, a personal computer system, or other electronic device. The computing system includes control circuitry and memory storing one or more sets of instructions. The one or more sets of instructions including instructions for performing any of the methods described herein. In some embodiments, the computing system includes an encoder component and a decoder component (e.g., a transcoder).

[0012] In accordance with some embodiments, a non-transitory computer-readable storage medium is provided. The non-transitory computer-readable storage medium stores one or more sets of instructions for execution by a computing system. The one or more sets of instructions including instructions for performing any of the methods described herein.

[0013] Thus, devices and systems are disclosed with methods for encoding and decoding video. Such methods, devices, and systems may complement or replace conventional methods, devices, and systems for video encoding/decoding.

[0014] The features and advantages described in the specification are not necessarily all-inclusive and, in particular, some additional features and advantages will be apparent to one of ordinary skill in the art in view of the drawings, specification, and claims provided in this disclosure. Moreover, it should be noted that the language used in the specification has been principally selected for readability and instructional purposes and has not necessarily been selected to delineate or circumscribe the subject matter described herein.

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] So that the present disclosure can be understood in greater detail, a more particular description can be had by reference to the features of various embodiments, some of which are illustrated in the appended drawings. The appended drawings, however, merely illustrate pertinent features of the present disclosure and are therefore not necessarily to be considered limiting, for the description can admit to other effective features as the person of skill in this art will appreciate upon reading this disclosure.

[0016] FIG. 1 is a block diagram illustrating an example communication system in accordance with some embodiments.

[0017] FIG. 2A is a block diagram illustrating example elements of an encoder component in accordance with some embodiments.

[0018] FIG. 2B is a block diagram illustrating example elements of a decoder component in accordance with some embodiments.

[0019] FIG. 3 is a block diagram illustrating an example server system in accordance with some embodiments.

[0020] FIG. 4A illustrates an example intra block copy technique in accordance with some embodiments.

[0021] FIG. 4B illustrates an example of generating a merge candidate list that includes a refined merge candidate in accordance with some embodiments.

[0022] FIG. 4C illustrates an example of generating a merge candidate list that includes a refined merge candidate in accordance with some embodiments.

[0023] FIG. 4D illustrates example predefined positions in a block in accordance with some embodiments.

[0024] FIG. 4E illustrates an example chained MV that includes BV and/or MV used to derive the refined merge candidate in accordance with some embodiments.

[0025] FIG. 4F illustrates an example of generating a merge candidate list that includes a refined merge candidate in accordance with some embodiments.

[0026] FIG. 5A illustrates an example video decoding process in accordance with some embodiments.

[0027] FIG. 5B illustrates an example video encoding process in accordance with some embodiments.

[0028] In accordance with common practice, the various features illustrated in the drawings are not necessarily drawn to scale, and like reference numerals can be used to denote like features throughout the specification and figures.

DETAILED DESCRIPTION

[0029] The present disclosure describes video/image compression techniques including using template matching to populate a merge candidate list that is used to reconstruct a current block. For example, a refined prediction block may be identified by applying a template-matching technique to a refinement search region associated with a block of an original merge candidate. A reference vector (e.g., a MV, a BV, and/or a DV) may be identified for the original merge candidate and a refined merge candidate list may be generated/populated for the current block using the reference vector. Afterwards, the current block may be reconstructed using information from the merge candidate list. In another example, a list of candidate predictors is generated for the current block by applying a template-matching technique to a block associated with an original merge candidate. In this example, the current block may be reconstructed using at least one candidate predictor from the list of candidate predictors.

[0030] Populating candidate lists for the current block based on a refined merge candidate identified using template matching may offer advantages including increasing coding accuracy while optionally reducing coding overhead. For example, coding accuracy may be increased by using one or more refined merge candidates that cover a wider range of coding cases, optionally while reducing a size of the merge candidate list. As an example, refined merge candidates identified using intra-template matching may more closely match the current block, instead of relying solely on neighboring blocks of the current block. Signaling overhead may be reduced when using the candidate list and signaling an index to the list instead of directly signaling MVs or other coding information. Coding efficiency may also be improved by populating the candidate list more efficiently, for example, based on template-matching costs, to provide better MV information for the current block.

Example Systems and Devices

[0031] FIG. 1 is a block diagram illustrating a communication system 100 in accordance with some embodiments. The communication system 100 includes a source device 102 and a plurality of electronic devices 120 (e.g., electronic device 120-1 to electronic device 120-m) that are communicatively coupled to one another via one or more networks. In some embodiments, the communication system 100 is a streaming system, e.g., for use with video-enabled applica-

tions such as video conferencing applications, digital TV applications, and media storage and/or distribution applications.

[0032] The source device **102** includes a video source **104** (e.g., a camera component or media storage) and an encoder component **106**. In some embodiments, the video source **104** is a digital camera (e.g., configured to create an uncompressed video sample stream). The encoder component **106** generates one or more encoded video bitstreams from the video stream. The video stream from the video source **104** may be high data volume as compared to the encoded video bitstream **108** generated by the encoder component **106**. Because the encoded video bitstream **108** is lower data volume (less data) as compared to the video stream from the video source, the encoded video bitstream **108** requires less bandwidth to transmit and less storage space to store as compared to the video stream from the video source **104**. In some embodiments, the source device **102** does not include the encoder component **106** (e.g., is configured to transmit uncompressed video to the network(s) **110**).

[0033] The one or more networks **110** represents any number of networks that convey information between the source device **102**, the server system **112**, and/or the electronic devices **120**, including for example wireline (wired) and/or wireless communication networks. The one or more networks **110** may exchange data in circuit-switched and/or packet-switched channels. Representative networks include telecommunications networks, local area networks, wide area networks and/or the Internet.

[0034] The one or more networks **110** include a server system **112** (e.g., a distributed/cloud computing system). In some embodiments, the server system **112** is, or includes, a streaming server (e.g., configured to store and/or distribute video content such as the encoded video stream from the source device **102**). The server system **112** includes a coder component **114** (e.g., configured to encode and/or decode video data). In some embodiments, the coder component **114** includes an encoder component and/or a decoder component. In various embodiments, the coder component **114** is instantiated as hardware, software, or a combination thereof. In some embodiments, the coder component **114** is configured to decode the encoded video bitstream **108** and re-encode the video data using a different encoding standard and/or methodology to generate encoded video data **116**. In some embodiments, the server system **112** is configured to generate multiple video formats and/or encodings from the encoded video bitstream **108**. In some embodiments, the server system **112** functions as a Media-Aware Network Element (MANE). For example, the server system **112** may be configured to prune the encoded video bitstream **108** for tailoring potentially different bitstreams to one or more of the electronic devices **120**. In some embodiments, a MANE is provided separate from the server system **112**.

[0035] The electronic device **120-1** includes a decoder component **122** and a display **124**. In some embodiments, the decoder component **122** is configured to decode the encoded video data **116** to generate an outgoing video stream that can be rendered on a display or other type of rendering device. In some embodiments, one or more of the electronic devices **120** does not include a display component (e.g., is communicatively coupled to an external display device and/or includes a media storage). In some embodiments, the electronic devices **120** are streaming clients. In

some embodiments, the electronic devices **120** are configured to access the server system **112** to obtain the encoded video data **116**.

[0036] The source device and/or the plurality of electronic devices **120** are sometimes referred to as “terminal devices” or “user devices.” In some embodiments, the source device **102** and/or one or more of the electronic devices **120** are instances of a server system, a personal computer, a portable device (e.g., a smartphone, tablet, or laptop), a wearable device, a video conferencing device, and/or other type of electronic device.

[0037] In example operation of the communication system **100**, the source device **102** transmits the encoded video bitstream **108** to the server system **112**. For example, the source device **102** may code a stream of pictures that are captured by the source device. The server system **112** receives the encoded video bitstream **108** and may decode and/or encode the encoded video bitstream **108** using the coder component **114**. For example, the server system **112** may apply an encoding to the video data that is more optimal for network transmission and/or storage. The server system **112** may transmit the encoded video data **116** (e.g., one or more coded video bitstreams) to one or more of the electronic devices **120**. Each electronic device **120** may decode the encoded video data **116** and optionally display the video pictures.

[0038] FIG. 2A is a block diagram illustrating example elements of the encoder component **106** in accordance with some embodiments. The encoder component **106** receives video data (e.g., a source video sequence) from the video source **104**. In some embodiments, the encoder component includes a receiver (e.g., a transceiver) component configured to receive the source video sequence. In some embodiments, the encoder component **106** receives a video sequence from a remote video source (e.g., a video source that is a component of a different device than the encoder component **106**). The video source **104** may provide the source video sequence in the form of a digital video sample stream that can be of any suitable bit depth (e.g., 8-bit, 10-bit, or 12-bit), any colorspace (e.g., BT.601 Y CrCb, or RGB), and any suitable sampling structure (e.g., Y CrCb 4:2:0 or Y CrCb 4:4:4). In some embodiments, the video source **104** is a storage device storing previously captured/prepared video. In some embodiments, the video source **104** is camera that captures local image information as a video sequence. Video data may be provided as a plurality of individual pictures that impart motion when viewed in sequence. The pictures themselves may be organized as a spatial array of pixels, where each pixel can include one or more samples depending on the sampling structure, color space, etc. in use. A person of ordinary skill in the art can readily understand the relationship between pixels and samples.

[0039] The encoder component **106** is configured to code and/or compress the pictures of the source video sequence into a coded video sequence **216** in real-time or under other time constraints as required by the application. In some embodiments, the encoder component **106** is configured to perform a conversion between the source video sequence and a bitstream of visual media data (e.g., a video bitstream). Enforcing appropriate coding speed is one function of a controller **204**. In some embodiments, the controller **204** controls other functional units as described below and is functionally coupled to the other functional units. Param-

eters set by the controller **204** may include rate-control-related parameters (e.g., picture skip, quantizer, and/or lambda value of rate-distortion optimization techniques), picture size, group of pictures (GOP) layout, maximum MV search range, and so forth. A person of ordinary skill in the art can readily identify other functions of controller **204** as they may pertain to the encoder component **106** being optimized for a certain system design.

[0040] In some embodiments, the encoder component **106** is configured to operate in a coding loop. In a simplified example, the coding loop includes a source coder **202** (e.g., responsible for creating symbols, such as a symbol stream, based on an input picture to be coded and reference picture (s)), and a (local) decoder **210**. The decoder **210** reconstructs the symbols to create the sample data in a similar manner as a (remote) decoder (when compression between symbols and coded video bitstream is lossless). The reconstructed sample stream (sample data) is input to the reference picture memory **208**. As the decoding of a symbol stream leads to bit-exact results independent of decoder location (local or remote), the content in the reference picture memory **208** is also bit exact between the local encoder and remote encoder. In this way, the prediction part of an encoder interprets as reference picture samples the same sample values as a decoder would interpret when using prediction during decoding.

[0041] The operation of the decoder **210** can be the same as of a remote decoder, such as the decoder component **122**, which is described in detail below in conjunction with FIG. 2B. Briefly referring to FIG. 2B, however, as symbols are available and encoding/decoding of symbols to a coded video sequence by an entropy coder **214** and the parser **254** can be lossless, the entropy decoding parts of the decoder component **122**, including the buffer memory **252** and the parser **254** may not be fully implemented in the local decoder **210**.

[0042] The decoder technology described herein, except the parsing/entropy decoding, may be to be present, in substantially identical functional form, in a corresponding encoder. For this reason, the disclosed subject matter focuses on decoder operation. Additionally, the description of encoder technologies can be abbreviated as they may be the inverse of the decoder technologies.

[0043] As part of its operation, the source coder **202** may perform motion compensated predictive coding, which codes an input frame predictively with reference to one or more previously-coded frames from the video sequence that were designated as reference frames. In this manner, the coding engine **212** codes differences between pixel blocks of an input frame and pixel blocks of reference frame(s) that may be selected as prediction reference(s) to the input frame. The controller **204** may manage coding operations of the source coder **202**, including, for example, setting of parameters and subgroup parameters used for encoding the video data.

[0044] The decoder **210** decodes coded video data of frames that may be designated as reference frames, based on symbols created by the source coder **202**. Operations of the coding engine **212** may advantageously be lossy processes. When the coded video data is decoded at a video decoder (not shown in FIG. 2A), the reconstructed video sequence may be a replica of the source video sequence with some errors. The decoder **210** replicates decoding processes that may be performed by a remote video decoder on reference

frames and may cause reconstructed reference frames to be stored in the reference picture memory **208**. In this manner, the encoder component **106** stores copies of reconstructed reference frames locally that have common content as the reconstructed reference frames that will be obtained by a remote video decoder (absent transmission errors).

[0045] The predictor **206** may perform prediction searches for the coding engine **212**. That is, for a new frame to be coded, the predictor **206** may search the reference picture memory **208** for sample data (as candidate reference pixel blocks) or certain metadata such as reference picture MVs, block shapes, and so on, that may serve as an appropriate prediction reference for the new pictures. The predictor **206** may operate on a sample block-by-pixel block basis to find appropriate prediction references. As determined by search results obtained by the predictor **206**, an input picture may have prediction references drawn from multiple reference pictures stored in the reference picture memory **208**.

[0046] Output of all aforementioned functional units may be subjected to entropy coding in the entropy coder **214**. The entropy coder **214** translates the symbols as generated by the various functional units into a coded video sequence, by losslessly compressing the symbols according to technologies known to a person of ordinary skill in the art (e.g., Huffman coding, variable length coding, and/or arithmetic coding).

[0047] In some embodiments, an output of the entropy coder **214** is coupled to a transmitter. The transmitter may be configured to buffer the coded video sequence(s) as created by the entropy coder **214** to prepare them for transmission via a communication channel **218**, which may be a hardware/software link to a storage device which would store the encoded video data. The transmitter may be configured to merge coded video data from the source coder **202** with other data to be transmitted, for example, coded audio data and/or ancillary data streams (sources not shown). In some embodiments, the transmitter may transmit additional data with the encoded video. The source coder **202** may include such data as part of the coded video sequence. Additional data may comprise temporal/spatial/SNR enhancement layers, other forms of redundant data such as redundant pictures and slices, Supplementary Enhancement Information (SEI) messages, Visual Usability Information (VUI) parameter set fragments, and the like.

[0048] The controller **204** may manage operation of the encoder component **106**. During coding, the controller **204** may assign to each coded picture a certain coded picture type, which may affect the coding techniques that are applied to the respective picture. For example, pictures may be assigned as an Intra Picture (I picture), a Predictive Picture (P picture), or a Bi-directionally Predictive Picture (B Picture). An Intra Picture may be coded and decoded without using any other frame in the sequence as a source of prediction. Some video codecs allow for different types of Intra pictures, including, for example Independent Decoder Refresh (IDR) Pictures. A person of ordinary skill in the art is aware of those variants of I pictures and their respective applications and features, and therefore they are not repeated here. A Predictive picture may be coded and decoded using intra prediction or inter prediction using at most one MV and reference index to predict the sample values of each block. A Bi-directionally Predictive Picture may be coded and decoded using intra prediction or inter prediction using at most two MVs and reference indices to predict the sample

values of each block. Similarly, multiple-predictive pictures can use more than two reference pictures and associated metadata for the reconstruction of a single block.

[0049] Source pictures commonly may be subdivided spatially into a plurality of sample blocks (for example, blocks of 4×4, 8×8, 4×8, or 16×16 samples each) and coded on a block-by-block basis. Blocks may be coded predictively with reference to other (already coded) blocks as determined by the coding assignment applied to the blocks' respective pictures. For example, blocks of I pictures may be coded non-predictively or they may be coded predictively with reference to already coded blocks of the same picture (spatial prediction or intra prediction). Pixel blocks of P pictures may be coded non-predictively, via spatial prediction or via temporal prediction with reference to one previously coded reference pictures. Blocks of B pictures may be coded non-predictively, via spatial prediction or via temporal prediction with reference to one or two previously coded reference pictures.

[0050] A video may be captured as a plurality of source pictures (video pictures) in a temporal sequence. Intra-picture prediction (often abbreviated to intra prediction) makes use of spatial correlation in a given picture, and inter-picture prediction makes uses of the (temporal or other) correlation between the pictures. In an example, a specific picture under encoding/decoding, which is referred to as a current picture, is partitioned into blocks. When a block in the current picture is similar to a reference block in a previously coded and still buffered reference picture in the video, the block in the current picture can be coded by a vector that is referred to as a motion vector. The MV points to the reference block in the reference picture, and can have a third dimension identifying the reference picture, in case multiple reference pictures are in use.

[0051] The encoder component 106 may perform coding operations according to a predetermined video coding technology or standard, such as any described herein. In its operation, the encoder component 106 may perform various compression operations, including predictive coding operations that exploit temporal and spatial redundancies in the input video sequence. The coded video data, therefore, may conform to a syntax specified by the video coding technology or standard being used.

[0052] FIG. 2B is a block diagram illustrating example elements of the decoder component 122 in accordance with some embodiments. The decoder component 122 in FIG. 2B is coupled to the channel 218 and the display 124. In some embodiments, the decoder component 122 includes a transmitter coupled to the loop filter 256 and configured to transmit data to the display 124 (e.g., via a wired or wireless connection).

[0053] In some embodiments, the decoder component 122 includes a receiver coupled to the channel 218 and configured to receive data from the channel 218 (e.g., via a wired or wireless connection). The receiver may be configured to receive one or more coded video sequences to be decoded by the decoder component 122. In some embodiments, the decoding of each coded video sequence is independent from other coded video sequences. Each coded video sequence may be received from the channel 218, which may be a hardware/software link to a storage device which stores the encoded video data. The receiver may receive the encoded video data with other data, for example, coded audio data and/or ancillary data streams, that may be forwarded to their

respective using entities (not depicted). The receiver may separate the coded video sequence from the other data. In some embodiments, the receiver receives additional (redundant) data with the encoded video. The additional data may be included as part of the coded video sequence(s). The additional data may be used by the decoder component 122 to decode the data and/or to more accurately reconstruct the original video data. Additional data can be in the form of, e.g., temporal, spatial, or SNR enhancement layers, redundant slices, redundant pictures, forward error correction codes, and so on.

[0054] In accordance with some embodiments, the decoder component 122 includes a buffer memory 252, a parser 254 (also sometimes referred to as an entropy decoder), a scaler/inverse transform unit 258, an intra picture prediction unit 262, a motion compensation prediction unit 260, an aggregator 268, the loop filter unit 256, a reference picture memory 266, and a current picture memory 264. In some embodiments, the decoder component 122 is implemented as an integrated circuit, a series of integrated circuits, and/or other electronic circuitry. The decoder component 122 may be implemented at least in part in software.

[0055] The buffer memory 252 is coupled in between the channel 218 and the parser 254 (e.g., to combat network jitter). In some embodiments, the buffer memory 252 is separate from the decoder component 122. In some embodiments, a separate buffer memory is provided between the output of the channel 218 and the decoder component 122. In some embodiments, a separate buffer memory is provided outside of the decoder component 122 (e.g., to combat network jitter) in addition to the buffer memory 252 inside the decoder component 122 (e.g., which is configured to handle playout timing). When receiving data from a store/forward device of sufficient bandwidth and controllability, or from an isosynchronous network, the buffer memory 252 may not be needed, or can be small. For use on best effort packet networks such as the Internet, the buffer memory 252 may be required, can be comparatively large and/or of adaptive size, and may at least partially be implemented in an operating system or similar elements outside of the decoder component 122.

[0056] The parser 254 is configured to reconstruct symbols 270 from the coded video sequence. The symbols may include, for example, information used to manage operation of the decoder component 122, and/or information to control a rendering device such as the display 124. The control information for the rendering device(s) may be in the form of, for example, Supplementary Enhancement Information (SEI) messages or Video Usability Information (VUI) parameter set fragments (not depicted). The parser 254 parses (entropy-decodes) the coded video sequence. The coding of the coded video sequence can be in accordance with a video coding technology or standard, and can follow principles well known to a person skilled in the art, including variable length coding, Huffman coding, arithmetic coding with or without context sensitivity, and so forth. The parser 254 may extract from the coded video sequence, a set of subgroup parameters for at least one of the subgroups of pixels in the video decoder, based upon at least one parameter corresponding to the group. Subgroups can include Groups of Pictures (GOPs), pictures, tiles, slices, macroblocks, Coding Units (CUs), blocks, Transform Units (TUs), Prediction Units (PUs) and so forth. The parser 254 may also

extract, from the coded video sequence, information such as transform coefficients, quantizer parameter values, MVs, and so forth.

[0057] Reconstruction of the symbols **270** can involve multiple different units depending on the type of the coded video picture or parts thereof (such as: inter and intra picture, inter and intra block), and other factors. Which units are involved, and how they are involved, can be controlled by the subgroup control information that was parsed from the coded video sequence by the parser **254**. The flow of such subgroup control information between the parser **254** and the multiple units below is not depicted for clarity.

[0058] The decoder component **122** can be conceptually subdivided into a number of functional units, and in some implementations, these units interact closely with each other and can, at least partly, be integrated into each other. However, for clarity, the conceptual subdivision of the functional units is maintained herein.

[0059] The scaler/inverse transform unit **258** receives quantized transform coefficients as well as control information (such as which transform to use, block size, quantization factor, and/or quantization scaling matrices) as symbol (s) **270** from the parser **254**. The scaler/inverse transform unit **258** can output blocks including sample values that can be input into the aggregator **268**. In some cases, the output samples of the scaler/inverse transform unit **258** pertain to an intra coded block; that is: a block that is not using predictive information from previously reconstructed pictures, but can use predictive information from previously reconstructed parts of the current picture. Such predictive information can be provided by the intra picture prediction unit **262**. The intra picture prediction unit **262** may generate a block of the same size and shape as the block under reconstruction, using surrounding already-reconstructed information fetched from the current (partly reconstructed) picture from the current picture memory **264**. The aggregator **268** may add, on a per sample basis, the prediction information the intra picture prediction unit **262** has generated to the output sample information as provided by the scaler/inverse transform unit **258**.

[0060] In other cases, the output samples of the scaler/inverse transform unit **258** pertain to an inter coded, and potentially motion-compensated, block. In such cases, the motion compensation prediction unit **260** can access the reference picture memory **266** to fetch samples used for prediction. After motion compensating the fetched samples in accordance with the symbols **270** pertaining to the block, these samples can be added by the aggregator **268** to the output of the scaler/inverse transform unit **258** (in this case called the residual samples or residual signal) so to generate output sample information. The addresses within the reference picture memory **266**, from which the motion compensation prediction unit **260** fetches prediction samples, may be controlled by MVs. The MVs may be available to the motion compensation prediction unit **260** in the form of symbols **270** that can have, for example, X, Y, and reference picture components. Motion compensation may also include interpolation of sample values as fetched from the reference picture memory **266**, e.g., when sub-sample exact MVs are in use, MV prediction mechanisms.

[0061] The output samples of the aggregator **268** can be subject to various loop filtering techniques in the loop filter unit **256**. Video compression technologies can include in-loop filter technologies that are controlled by parameters

included in the coded video bitstream and made available to the loop filter unit **256** as symbols **270** from the parser **254**, but can also be responsive to meta-information obtained during the decoding of previous (in decoding order) parts of the coded picture or coded video sequence, as well as responsive to previously reconstructed and loop-filtered sample values. The output of the loop filter unit **256** can be a sample stream that can be output to a render device such as the display **124**, as well as stored in the reference picture memory **266** for use in future inter-picture prediction.

[0062] Certain coded pictures, once reconstructed, can be used as reference pictures for future prediction. Once a coded picture is reconstructed and the coded picture has been identified as a reference picture (by, for example, parser **254**), the current reference picture can become part of the reference picture memory **266**, and a fresh current picture memory can be reallocated before commencing the reconstruction of the following coded picture.

[0063] The decoder component **122** may perform decoding operations according to a predetermined video compression technology that may be documented in a standard, such as any of the standards described herein. The coded video sequence may conform to a syntax specified by the video compression technology or standard being used, in the sense that it adheres to the syntax of the video compression technology or standard, as specified in the video compression technology document or standard and specifically in the profiles document therein. Also, for compliance with some video compression technologies or standards, the complexity of the coded video sequence may be within bounds as defined by the level of the video compression technology or standard. In some cases, levels restrict the maximum picture size, maximum frame rate, maximum reconstruction sample rate (measured in, for example megasamples per second), maximum reference picture size, and so on. Limits set by levels can, in some cases, be further restricted through Hypothetical Reference Decoder (HRD) specifications and metadata for HRD buffer management signaled in the coded video sequence.

[0064] FIG. 3 is a block diagram illustrating the server system **112** in accordance with some embodiments. The server system **112** includes control circuitry **302**, one or more network interfaces **304**, a memory **314**, a user interface **306**, and one or more communication buses **312** for inter-connecting these components. In some embodiments, the control circuitry **302** includes one or more processors (e.g., a CPU, GPU, and/or DPU). In some embodiments, the control circuitry includes field-programmable gate array(s), hardware accelerators, and/or integrated circuit(s) (e.g., an application-specific integrated circuit).

[0065] The network interface(s) **304** may be configured to interface with one or more communication networks (e.g., wireless, wireline, and/or optical networks). The communication networks can be local, wide-area, metropolitan, vehicular and industrial, real-time, delay-tolerant, and so on. Examples of communication networks include local area networks such as Ethernet, wireless LANs, cellular networks to include GSM, 3G, 4G, 5G, LTE and the like, TV wireline or wireless wide area digital networks to include cable TV, satellite TV, and terrestrial broadcast TV, vehicular and industrial to include CANBus, and so forth. Such communication can be unidirectional, receive only (e.g., broadcast TV), unidirectional send-only (e.g., CANbus to certain CANbus devices), or bi-directional (e.g., to other computer

systems using local or wide area digital networks). Such communication can include communication to one or more cloud computing networks.

[0066] The user interface 306 includes one or more output devices 308 and/or one or more input devices 310. The input device(s) 310 may include one or more of: a keyboard, a mouse, a trackpad, a touch screen, a data-glove, a joystick, a microphone, a scanner, a camera, or the like. The output device(s) 308 may include one or more of: an audio output device (e.g., a speaker), a visual output device (e.g., a display or monitor), or the like.

[0067] The memory 314 may include high-speed random-access memory (such as DRAM, SRAM, DDR RAM, and/or other random access solid-state memory devices) and/or non-volatile memory (such as one or more magnetic disk storage devices, optical disk storage devices, flash memory devices, and/or other non-volatile solid-state storage devices). The memory 314 optionally includes one or more storage devices remotely located from the control circuitry 302. The memory 314, or, alternatively, the non-volatile solid-state memory device(s) within the memory 314, includes a non-transitory computer-readable storage medium. In some embodiments, the memory 314, or the non-transitory computer-readable storage medium of the memory 314, stores the following programs, modules, instructions, and data structures, or a subset or superset thereof:

- [0068] an operating system 316 that includes procedures for handling various basic system services and for performing hardware-dependent tasks;
- [0069] a network communication module 318 that is used for connecting the server system 112 to other computing devices via the one or more network interfaces 304 (e.g., via wired and/or wireless connections);
- [0070] a coding module 320 for performing various functions with respect to encoding and/or decoding data, such as video data. In some embodiments, the coding module 320 is an instance of the coder component 114. The coding module 320 including, but not limited to, one or more of:
- [0071] a decoding module 322 for performing various functions with respect to decoding encoded data, such as those described previously with respect to the decoder component 122; and
- [0072] an encoding module 340 for performing various functions with respect to encoding data, such as those described previously with respect to the encoder component 106; and
- [0073] a picture memory 352 for storing pictures and picture data, e.g., for use with the coding module 320. In some embodiments, the picture memory 352 includes one or more of: the reference picture memory 208, the buffer memory 252, the current picture memory 264, and the reference picture memory 266.
- [0074] In some embodiments, the decoding module 322 includes a parsing module 324 (e.g., configured to perform the various functions described previously with respect to the parser 254), a transform module 326 (e.g., configured to perform the various functions described previously with respect to the scalar/inverse transform unit 258), a prediction module 328 (e.g., configured to perform the various functions described previously with respect to the motion compensation prediction unit 260 and/or the intra picture prediction unit 262), and a filter module 330 (e.g., configured to

perform the various functions described previously with respect to the loop filter 256).

[0075] In some embodiments, the encoding module 340 includes a code module 342 (e.g., configured to perform the various functions described previously with respect to the source coder 202 and/or the coding engine 212) and a prediction module 344 (e.g., configured to perform the various functions described previously with respect to the predictor 206). In some embodiments, the decoding module 322 and/or the encoding module 340 include a subset of the modules shown in FIG. 3. For example, a shared prediction module is used by both the decoding module 322 and the encoding module 340.

[0076] Each of the above identified modules stored in the memory 314 corresponds to a set of instructions for performing a function described herein. The above identified modules (e.g., sets of instructions) need not be implemented as separate software programs, procedures, or modules, and thus various subsets of these modules may be combined or otherwise re-arranged in various embodiments. For example, the coding module 320 optionally does not include separate decoding and encoding modules, but rather uses a same set of modules for performing both sets of functions. In some embodiments, the memory 314 stores a subset of the modules and data structures identified above. In some embodiments, the memory 314 stores additional modules and data structures not described above.

[0077] Although FIG. 3 illustrates the server system 112 in accordance with some embodiments, FIG. 3 is intended more as a functional description of the various features that may be present in one or more server systems rather than a structural schematic of the embodiments described herein. In practice, items shown separately could be combined and some items could be separated. For example, some items shown separately in FIG. 3 could be implemented on single servers and single items could be implemented by one or more servers. The actual number of servers used to implement the server system 112, and how features are allocated among them, will vary from one implementation to another and, optionally, depends in part on the amount of data traffic that the server system handles during peak usage periods as well as during average usage periods.

Example Coding Techniques

[0078] The coding processes and techniques described below may be performed at the devices and systems described above (e.g., the source device 102, the server system 112, and/or the electronic device 120). According to some embodiments, example methods for using template matching techniques to populate candidate lists (e.g., used to reconstruct a current blocks) are described below.

[0079] As described above, in general, a hybrid video codec includes the following coding modules, intra prediction, inter prediction, transform coding, quantization, entropy coding, and post in-loop filter. In inter prediction coding, a final motion vector may be derived based on spatial/temporal information or be the sum of a signalled motion vector difference and a derived or selected motion vector predictor. When a final motion vector is derived based on spatial or temporal information, the motion vector can be derived from the motion vector of adjacent neighbouring coded block, non-adjacent neighbouring coded block, collocated coded block in the reference picture, or history-based motion information from the previous coded block.

The positions of those coded blocks are fixed and predefined during the merge candidate list construction.

[0080] As is known to those of skill in the art, an intra block copy technique is a technique that identifies a prediction block for the current block using a BV. The BV is used to identify another block in the same picture of the current block that may be adjacent or non-adjacent to the current block. The BV may be either explicitly signaled or implicitly derived. When the BV is explicitly signaled, it is usually referred as an intra block copy (IBC) method. When the BV is implicitly derived such as by comparing a template area (a group of neighboring reconstruction samples located adjacent to the block) between the current block and a candidate prediction block, it is usually referred as an intra template matching method or a template-based intra mode derivation (TIMD) method.

[0081] In some instances, a current coding block and its neighboring samples share a similar texture characteristic. In such scenarios, the neighboring reconstructed samples of a current block, collectively called “a template,” can be employed to predict the current block. Template-matching may be used in inter prediction to derive the prediction block by calculating a distortion between the template of the current block and the template of the prediction block in the reference picture. Template-matching can also be applied in intra prediction, termed “intra template-matching,” on the reconstructed area of the current picture.

[0082] A “superblock size” or “coding tree unit (CTU)” may refer to the largest coding block size applied for coding an image/video picture or a video sequence. Additionally, block size (or region size) may refer to the block/region width, height, area, number of samples in the block/region, a max (or min) between block/region width and height, and/or block/region aspect ratio.

[0083] FIG. 4A illustrates an example of an intra block copy technique in accordance with some embodiments. In this example, the intra block copy technique includes identifying, via a predicted BV 408, a prediction block 404 within the same picture 400 as a current block 402. FIG. 4A shows an example in which the prediction block 404 is non-adjacent to the current block 402. In some embodiments, the prediction block 404 is adjacent to the current block 402. In some embodiments, the prediction BV 408 is selected from a list of candidate BVs. For example, the list of candidate BVs may be populated by BVs used in neighboring blocks and/or BVs from a BV bank. A BV predictor index may be signaled to indicate which candidate BV in the candidate list is used to predict the BV for the current block 402.

[0084] A block vector difference (BVD) 410 represents the difference between the predicted BV and an actual BV 406, which is the vector between corresponding portions of the current block (e.g., current block 402) and the prediction block (e.g., prediction block 404). While the BVD 410 depicted in FIG. 4A has components along both the horizontal and vertical dimensions, a BVD may extend along a single dimension or span two or more dimensions. In some embodiments, the BVD 410 includes information representing a magnitude of a BVD and/or a direction of the BVD, one or both of which may be signaled in the bitstream as intra block copy information or syntaxes.

[0085] Neighboring reconstructed samples of a current block, collectively called “a template,” may be used to predict the current block. In some embodiments, a template-

matching process is applied to a current picture to find at least one adjacent and/or non-adjacent block having MV and/or BV information usable for the current block. The MV and/or BV from that adjacent block and/or non-adjacent block(s) may be used to construct a merge candidate list for the current block. FIG. 4B illustrates an example of generating a merge candidate list that includes a refined merge candidate in accordance with some embodiments. In FIG. 4B, a current picture 420 includes a current block 412. A block 416 is selected from a merge candidate list, and a template matching process (e.g., intra-template matching) refinement process is used to identify a BV 421 of the block 416, in accordance with some embodiments. For example, the block 416 has a template 414 that includes a number of reconstructed samples. A distortion between the template 414 and other templates within a refinement search region 422 of the current picture 420 (e.g., templates formed by reconstructed samples in the refinement search region 422) is calculated, and a prediction block 418 may be identified, which is associated with a template 419 having a small or smallest distortion, and/or the lowest template-matching cost (e.g., based on a sum of absolute difference (SAD), a sum of absolute transformed differences (SATD), a sum of squared error (SSE), or another metric), thereby identifying the BV 421. In the example of FIG. 4B, the predicted block 418 has a MV 424, also denoted as MV_A , that points to a reference block 426 in a reference picture 428. A refined merge candidate 431 (e.g., also denoted as an MV) is derivable by either adding the BV 421 and the MV 424 (e.g., $MV = MV_A + BV$), or setting the refined merge candidate 431 as equal to the MV 424 (e.g., $MV = MV_A$). For example, when the refined merge candidate 431 is set as the MV 424, the final refined merge candidate 431 does not account for the non-adjacency of the prediction block 418 to the block 416 or the non-adjacency of the block 416 to the current block 412 (e.g., the displacement, via BV, of the prediction block 418 from the current block 416, is ignored, and/or a displacement via a displacement vector described below with reference to FIG. 4C between the block 416 and the current block 412 is ignored).

[0086] FIG. 4C illustrates an example of generating a merge candidate list that includes a refined merge candidate in accordance with some embodiments. The same reference numerals are used in FIG. 4C to refer to elements analogous to those illustrated in FIG. 4B. A displacement vector DV 442 corresponding to the displacement between the current block 412 and the block 416, which is used to search for the prediction block 418 associated with the refined merge candidate is also used to construct the final merge candidate. For example, the derived MV at that refined merge candidate is equal to $MV_A + BV + DV$ during the merge list construction.

[0087] In some embodiments, the refined merge candidate 431 or the refined merge candidate 444 (e.g., MV_A) is derived from a predefined motion field within prediction block 418 having dimensions $W \times H$. For example, the prediction block 418 (e.g., 32×32 , same size as the current block 412) may include multiple subblocks (e.g., each subblock having a size of 4×4 , resulting in an arrangement of 8×8 different subblocks, containing 64 MVs). For example, as illustrated in FIG. 4D, the prediction block 418 in FIG. 4B may correspond to a larger block 430 (e.g., sometimes referred to as a “predefined motion field”) in FIG. 4D, which includes various subblocks, such as a top right

subblock **432**, a lower right subblock **434**, a lower left subblock **436**, a top left subblock **438** and a center subblock **440**. In some embodiments, the availability of a MV within the motion field is checked by scanning the availability of MVs at one or more positions, optionally in a predefined scanning order when multiple positions are checked. In some embodiments, for example, the center subblock **440** (e.g., at a position $[W/2, H/2]$) is first checked to determine if MV information is available. In some embodiments, in accordance with a determination that MV information is not available at the center subblock **440**, MV information is checked at the top right subblock **432**, the lower right subblock **434**, the lower left subblock **436**, and the top left subblock **438** in sequence to determine whether MV information is available or not. If no MV at center position is available (e.g., the subblock **440** is intra coded), the MV information at the four corners (e.g., intra mode of the top left subblock **438**, the top right subblock **432**, the lower right subblock **434**, and the lower left subblock **436** are determined in a clockwise order, or using a different order). In some embodiments, if no MV is available at those predefined locations (center, top left, top right, lower right, lower left, the subblocks at the predefined locations are intra-coded), no refined merge candidate **431** or no refined merge candidate **444** (e.g., $MV_{A'}$) is available to the current block **412** (e.g., other subblocks in the non-shaded portions of the block **430** are not checked).

[0088] A distortion or template-matching cost between the template **414** and other templates within the refinement search region **422** of the current picture **420** may be calculated to select a template having a low (e.g., the lowest) template-matching cost. For example, selecting at template having an associated cost that is below a predetermined threshold. In some embodiments, the template-matching cost may be based on an SAD, SATD, SSE, or another metric.

[0089] In some embodiments, the prediction block is searched within a smaller predefined search area within a reconstruction area of the current picture **420**. For example, a search range restriction can be applied to search within a search range of a fixed size, to search within a current CTU row, to search within the current CTU row and/or to search within the N previously coded CTU row(s), etc. In some embodiments, instead of finding a single prediction block **418** (e.g., within the current picture **420**), additional prediction blocks (e.g., N different prediction blocks) having the lowest N template matching costs are searched within the current picture **420** and the respective MVs within these N blocks are derived.

[0090] In some embodiments, instead of the L-shaped templates (e.g., template **414** and the template **419**) shown in FIG. 4B, a template of a different shape or dimension is used for intra template-matching. For example, only a left template (e.g., left portion of the L-shaped template **414**), or only a top template (e.g., the horizontal portion of the L-shaped template **414**), or a top-left template (e.g., the L-shaped template **414**) may be selected/used. For example, for smaller blocks (e.g. blocks smaller than or equal a size threshold such as 64, 32, or a different value), a template having two lines of reconstructed samples is used while for blocks larger than the size threshold (e.g., larger than 64, 32, or a different value), a template having four lines of reconstructed samples is used. Thus, the template size may be dependent on the block size. In some embodiments, the

templates having different template-matching types and shapes are adaptively selected at the block level (e.g., using any suitable method to determine which template type is the best at the time of coding the current block) and, optionally, a syntax is signaled into the bitstream by the encoder to indicate which template type and/or shape is used.

[0091] In some embodiments, pixel subsampling within the template is used in the template-matching process to calculate the template-matching cost. For example, for a template of 32 pixels (e.g., 16×2 size), instead of calculating a pixel difference (e.g., an absolute difference) for each of the 32 pixels (e.g., all samples within the template), pixel differences are only calculated partially, for example, for just the even positions or checkerboard positions. Pixel subsampling allows a reduction in computation time and may help with hardware designs. In some embodiments, the template matching process is performed at a coarser step. For example, the step of search is changed to two samples per iteration instead of searching every sample (e.g., step size of one). For example, instead of searching every position in a 32×32 block, search is conducted at only even or only odd positions, so that the search is only conducted in a partial region.

[0092] In some embodiments, the refined merge candidate (e.g., refined merge candidate **431** or refined merge candidate **444**) is inserted into the merge candidate list before merge candidates derived from non-adjacent neighboring blocks and/or history-based motion information (e.g., MVs from previous blocks that are placed into a data buffer, such as a first in first out (FIFO) buffer). For example, merge candidates generated from prediction blocks (e.g., obtained through intra-template matching) may be closer and/or more similar to the current block, and may be more accurate than merge candidates constructed from other non-adjacent neighboring blocks.

[0093] FIG. 4E illustrates an example chained MV that includes BV and/or MV used to derive the refined merge candidate in accordance with some embodiments. In FIG. 4E, a current picture **443** includes a current block **446**. In some embodiments, a block **447** is selected from a merge candidate list, and a template matching process (e.g., intra-template matching) refinement process is used to identify a BV **452** (e.g., also denoted as BV) of the block **447**. The block **447** has a template **454**, and a distortion between the template **454** and other templates within a refinement search region **456** of the current picture **443** (e.g., templates formed by reconstructed samples in the refinement search region **456** of the current picture **443**) may be calculated, and a prediction block **448** associated with a template having the smallest distortion, and/or the lowest template-matching cost (e.g., based on SAD, SATD, SSE, or another metric) is identified, analogous to the identification of the BV **421** of the block **416** illustrated in FIG. 4B. In contrast to the prediction block **416** in FIG. 4B which has an associated MV **424**, another BV **458**, also denoted as $BV_{A' \rightarrow A''}$, is derived within the BV field of the prediction block **448** in FIG. 4D. The BV **458** $BV_{A' \rightarrow A''}$ pointed to another prediction block **460** from the prediction block **448**. The predicted block **460** has a MV **461**, also denoted as $MV_{A''}$, that points to a reference block **462** in a reference picture **464**. A refined final merge MV, or a final merge candidate, **466** may be equal to $MV_{A''}$. FIG. 4F illustrates an example of generating a merge candidate list that includes a refined merge candidate in accordance with some embodiments. The same

reference numerals are used in FIG. 4F to refer to elements analogous to those illustrated in FIG. 4E. A displacement vector DV 470 corresponding to the displacement between the current block 446 and the block 447, which is used to search for the prediction block 448 associated with the refined merge candidate is also used to construct the final merge candidate. For example, the derived MV at the refined merge candidate with the chain MV information is equal to $MV_{A'} + BV_{A' \rightarrow A''} + BV + DV$ (e.g., the derived MV is a chained MV that is derived by adding the BV 452 (e.g., BV), the BV 458 (e.g., $BV_{A' \rightarrow A''}$), the DV 470 and the MV 461 (e.g., $MV_{A''}$). In some embodiments, the depth of the chained MV propagation is limited. In such scenarios, for example, the depth of the chained MV propagation depicted in FIG. 4E would be two. In some embodiments, adding a BV is not considered to increase the depth of the chained MV propagation. In such scenarios, for example, the depth of the chained MV propagation depicted in FIG. 4E would be one (or zero). In some embodiments, a flag is signaled in high-level syntax, such as SPS, PPS, APS, picture header, or slice header, to indicate whether a BV is used for chained MV construction.

[0094] In some embodiments, the chained MV construction is applied with a clipping operation to ensure that the MV (e.g., MV 461) remains in a predefined area (e.g., a central portion, or in any other specific area) in the reference picture (e.g., reference picture 464).

[0095] In some embodiments, instead of searching within a refinement search region (e.g., refinement search region 422 or refinement search region 456), a refined merge candidate is identified by applying a displacement vector between a current block and the merge candidate to construct the merged MV with this displacement information. For example, the final merge MV is equal to the MV information derived from the merge candidate and the displacement vector. In some embodiments, the merge candidate may be any kind of a spatial merge candidate, such as a non-adjacent spatial merge candidate, an adjacent spatial merge candidate, or a pairwise merge candidate that includes adding two spatial vectors together, and/or a merge candidate that takes spatial history into account. In some embodiments, the displacement vector is derived based on a displacement offset between the current block position and the merge candidate position.

[0096] In some embodiments, the candidate list employs a two-step strategy. In a first step, a candidate list is built with a template matching constraint to contain candidates only within the current picture. In a second step, the first M ($M \leq N$) candidates in the list are checked to determine if the chained MV_i' (e.g., $MV_i' = MV_i + BV_i$, where i is between 1 to M) constitutes a new candidate. Any new candidate motion vectors MV_i' (up to M numbers of new candidate MVs) are inserted to the list of candidates from within the current picture only. For example, the insertion of candidates constructed using chained MV is appended in the candidate list, and a larger candidate list with L candidates is built, where $L > N$. In some embodiments, the insertion of new candidates constructed using chained MV is based on template costs and the list size is kept unchanged. For example, a highest template cost chained MV is removed from the list to include a chained MV having a lower template cost.

[0097] FIG. 5A is a flow diagram illustrating a method 500 of decoding video in accordance with some embodiments. The method 500 may be performed at a computing system

(e.g., the server system 112, the source device 102, or the electronic device 120) having control circuitry and memory storing instructions for execution by the control circuitry. In some embodiments, the method 500 is performed by executing instructions stored in the memory (e.g., the memory 314) of the computing system.

[0098] The system receives (502) a video bitstream (e.g., a coded video sequence) comprising a current picture composed of a plurality of blocks, including a current block. The system identifies (504) a merge candidate for the current block, identifies (506) a refined merge candidate by refining a position of the merge candidate using a template-matching technique, and identifies (508) a reference vector for the refined merge candidate. The system populates (510) a merge candidate list for the current block using the reference vector and reconstructs (512) the current block using information from the merge candidate list. In this way, a template-matching process may be applied in a current reconstructed picture within a predefined search window to refine a position of a merge candidate. After the refinement, a motion vector is derived on that refined merge candidate to construct the merge candidate list.

[0099] In some embodiments, the refined merge candidate may be derived from the adjacent neighboring block, non-adjacent neighboring block, collocated block from the collocated picture, history-based motion information from the previous coded block, and the like. In some embodiments, a flag is signaled in high-level syntax to indicate whether the refined merge candidate described herein is applied on the merge candidate. If the flag indicates usage of the refined merge candidate, the refinement procedure described herein is applied on the merge candidate. Otherwise, the original derived merge candidate is used for the merge list construction. The high-level syntax may be SPS, PPS, slice header, picture header, and the like.

[0100] In some embodiments, a flag is signaled to indicate whether the refined merge candidate is constructed in the merge list or not. If the flag indicates usage of the refined merge candidate, the refined merge candidate described herein is constructed in the merge list.

[0101] In some embodiments, the refined merge candidate described herein is inserted right after or right before the merge candidate obtained without the described refinement procedure. In some embodiments, the refined merge candidate described herein is constructed in the merge list before or after the original merge list construction.

[0102] In some embodiments, a threshold value is applied to determine whether the refined merge candidate is constructed in the merge list or not. The refined merge candidate may be constructed in the merge list if the template-matching cost is smaller than and/or equal to the threshold value. In some embodiments, the threshold value may be a predefined value. In some embodiments, the threshold value may be a QP dependent predefined value, such that this threshold value can be changed according to the block QP. In some embodiments, the threshold value may be a block size dependent value. In some embodiments, the threshold value may be a coding mode dependent value.

[0103] In some embodiments, an adaptive selection between a refined merge candidate and a non-refined merge candidate (which is derived using the original method) is applied to adaptively select one of the two candidates to be used to construct the merge candidate. The refined merge candidate is selected to construct the merge list when the

template-matching cost is smaller than and/or equal to the threshold value. Otherwise, the non-refined merge candidate is selected for the merge list construction. In some embodiments, the threshold value derivation described above is also applicable here.

[0104] In some embodiments, the MV is derived based on the motion information derivation of the refined merge candidate. An example is shown in FIG. 4B. The merge candidate A is derived using the merge list construction procedure. A coded block A' with MV information $MV_{A'}$ which has the smallest template-matching cost between block C and block A' is derived by using template-matching process within a predefined search window in the current reconstructed picture, and the derived MV information in block A' points to a reference picture of the current block C. Therefore, the derived MV of the refined merge candidate by using intra template-matching is equal to $MV_{A'}$.

[0105] In some embodiments, the MV is derived based on the motion information derivation of the refined merge candidate. An example is shown as follows in FIG. 4D. The merge candidate A is derived by using the merge list construction procedure. A coded block A' with MV information $MV_{A'}$ which has the smallest template-matching cost between block C and block A' is derived using a template-matching process within a predefined search window in the current reconstructed picture, and the derived MV information in block A' points to the reference picture that is used for the current block C. Moreover, the displacement between the current block and the refined merge candidate is also used to construct the final merge candidate. Finally, the derived motion vector at that refined merge candidate is equal to $MV_{A'} + BV + DV$ during the merge list construction.

[0106] In some embodiments, a flag is signaled in high level syntax such as SPS, PPS, APS, picture header, slice header, to indicate whether the block vector is used for lookahead MV predictor construction.

[0107] In some embodiments, the derived $MV_{A'}$ is derived from a predefined motion field within the $W \times H$ block A', and the block size of block A' is equal to the current block size. For example, the motion field may be 4×4 . The block A' (e.g., prediction block 418) may be 16×16 , and the motion information in the 4×4 grid may include some portions that are intra coded, some portions that are inter coded, and some portions may not have motion information. The availability of a MV within the motion field is checked by scanning the availability of MV at no less than one position and this scanning order is a predefined order if multiple positions are checked. FIG. 4D shows an example of the MV derivation. The system determines if a MV is available or not at a center position $[W/2, H/2]$. If no MV at the center position is available, the MV information at four corners is determined in sequence.

[0108] In some embodiments, the template-matching cost is SAD, SATD, SSE, or the like. In some embodiments, a search area of the intra template-matching is within the reconstructed current picture. In some embodiments, a search range restriction is applied to the intra template-matching, for example, within a search range with a fixed size, within a current CTU row, within the current CTU row and N previously coded CTU row(s), etc.

[0109] In some embodiments, different template-matching types are used for intra template-matching. In some embodiments, only a left template, a top template, or a top-left template is used. In some embodiments, for smaller blocks

(e.g. smaller than or equal to 64), the template size may be 2 lines while for other cases (e.g., larger blocks), the template size may be 4 lines. In some embodiments, these different template-matching types can be used adaptively at the block level and a syntax is signaled to indicate which template type is used.

[0110] In some embodiments, pixel subsampling within the template can be used during the template-matching process to calculate the template-matching cost. In some embodiments, the template matching process is performed at a coarser step. For example, the step of search is changed to two samples per iteration instead of one.

[0111] In some embodiments, a chained MV which includes a BV and/or a MV can be used to derive the merge candidate by using intra template-matching. FIG. 4D and FIG. 4E show two examples of merge MV derivation with chained MV and/or BV by using intra template-matching. In FIG. 4D, a block A' is derived using the intra template-matching process. The associated $BV_{A' \rightarrow A''}$ is derived from the corresponding BV field within block A', and the block vector $BV_{A' \rightarrow A''}$ points to block A". Another motion vector, $MV_{A''}$, is derived from block A" for the final merged MV derivation. In the first example in FIG. 4D, the final merged MV is equal to the MV derived from the block A". The second example in FIG. 4E shows how to derive the final merge MV with the chained MV information.

[0112] In some embodiments, the depth of chained MV propagation is limited. In some embodiments, adding BV is not considered to increase the depth of the chained MV propagation.

[0113] In some embodiments, a flag is signaled in high level syntax such as SPS, PPS, APS, picture header, slice header, to indicate whether the BV is used for chained MV predictor construction or not.

[0114] In some embodiments, the chained MV methodologies and systems described herein are applied with a clipping operation, e.g., in order to limit the MV to within a predefined restricted area in the reference picture.

[0115] In some embodiments, the size of the merge list is changed when the refined merge candidate is used. In some embodiments, the size of merge list is increased by a fixed size N, where N additional refined merge candidates are allowed to be inserted into the merge candidate. In some embodiments, the size of merge list is adaptively determined. If no better candidate is found other than the starting candidate A for all candidates in the merge list, the size of merge list is not changed. Otherwise, the size of merge list is increased by a number of M candidates, which are M better refined candidates compared to their starting candidates.

[0116] The disclosed methodologies and systems include applying a displacement vector between a current block and the merge candidate to construct the merged MV with this displacement information. More specifically, the final merge MV is equal to the MV information derived from the merge candidate and the displacement vector. In some embodiments, the merge candidate may be any kind of a spatial merge candidate. In some embodiments, the displacement vector is derived based on a displacement offset between the current block position and the merge candidate position.

[0117] FIG. 5B is a flow diagram illustrating a method 550 of encoding video in accordance with some embodiments. The method 550 may be performed at a computing system (e.g., the server system 112, the source device 102, or the

electronic device **120**) having control circuitry and memory storing instructions for execution by the control circuitry. In some embodiments, the method **550** is performed by executing instructions stored in the memory (e.g., the memory **314**) of the computing system. In some embodiments, the method **550** is performed by a same system as the method **500** described above.

[0118] The system receives (**552**) video data (e.g., a source video sequence) comprising a current picture composed of a plurality of blocks, including a current block. The system identifies (**552**) a merge candidate for the current block and identifies (**554**) a refined merge candidate by refining a position of the merge candidate using a template-matching technique. The system identifies (**556**) a reference vector for the refined merge candidate, populates (**558**) a merge candidate list for the current block using the reference vector and encodes (**560**) the current block using information from the merge candidate list. As described previously, the encoding process may mirror the decoding processes described herein (e.g., template matching and merge candidate list construction as described above). For brevity, those details are not repeated here.

[0119] Although FIGS. **5A** and **5B** illustrate a number of logical stages in a particular order, stages which are not order dependent may be reordered and other stages may be combined or broken out. Some reordering or other groupings not specifically mentioned will be apparent to those of ordinary skill in the art, so the ordering and groupings presented herein are not exhaustive. Moreover, it should be recognized that the stages could be implemented in hardware, firmware, software, or any combination thereof.

[0120] Turning now to some example embodiments.

[0121] (A1) In one aspect, some embodiments include a method (e.g., the method **500**) of video decoding. In some embodiments, the method is performed at a computing system (e.g., the server system **112**) having memory and control circuitry. In some embodiments, the method is performed at a coding module (e.g., the coding module **320**). In some embodiments, the method is performed at a source coding component (e.g., the source coder **202**), a coding engine (e.g., the coding engine **212**), and/or an entropy coder (e.g., the entropy coder **214**). The method includes (i) receiving a video bitstream (e.g., a coded video sequence) comprising a current picture composed of a plurality of blocks, including a current block and (ii) identifying a merge candidate for the current block. The method also includes (iii) identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique, (iv) identifying a reference vector for the refined merge candidate, (v) populating a merge candidate list for the current block using the reference vector; and (vi) reconstructing the current block using information from the merge candidate list. For example, a template-matching processing may be applied in the current reconstructed picture within a predefined search window to refine the position of the merge candidate. After the position refinement, a motion vector is derived on that refined merge candidate to construct the merge candidate list. The merge candidate list is sometimes simply referred to as a merge list.

[0122] (A2) In some embodiments of A1, the refined merge candidate comprises an adjacent or non-adjacent spatial neighboring block to the current block. For example, the refined merge candidate may be derived from an adjacent neighboring block, non-adjacent neighboring block, co-

located block from a co-located picture, history-based motion information from the previous coded block, and the like.

[0123] (A3) In some embodiments of A1 or A2, the refined merge candidate comprises a block co-located with the current block in a reference picture. In some embodiments, the refined merge candidate comprises history-based motion information from a previously-coded block.

[0124] (A4) In some embodiments of any of A1-A3, the method further comprises parsing an indicator in the video bitstream, the indicator indicating whether merge candidate refinement is enabled for the current block, wherein the refined merge candidate is identified when the indicator indicates that merge candidate refinement is enabled for the current block. For example, a flag may be signaled in high-level syntax to indicate whether the refined merge candidate is applied on the merge candidate or not. As an example, if the flag is true, the refined procedure is applied on the merge candidate; and otherwise, the original derived merge candidate is applied on the merge list construction. For example, the high-level syntax can be SPS, PPS, slice header, picture header, or other high-level syntax. In some embodiments, a size of the merge candidate list is increased when merge candidate refinement is enabled. For example, the size of merge candidate list may be increased by a fixed size N, where N additional refined merge candidates are allowed to be inserted into the merge candidate. In some embodiments, the size of the merge candidate list is based on whether a viable refined merge candidate is identified. For example, the size of the merge candidate list may be based on whether a refined merge candidate is identified with a template-matching cost that is less than a threshold. As an example, the size of merge list may be adaptively determined. In this example, if no better candidate could be found other than the starting candidate A for all candidates in the merge list, the size of merge list is not changed. Otherwise, the size of merge list is increased by a number of M candidates, where M better refined candidates are found (respectively compared to their starting candidates).

[0125] (A5) In some embodiments of any of A1-A4, the method further comprises parsing an indicator in the video bitstream, the indicator indicating whether the refined merge candidate is to be included in the merge candidate list. For example, a flag is signaled to indicate whether the refined merge candidate is constructed in the merge list or not. If the flag is true, the proposed refined merge candidate is constructed in the merge list.

[0126] (A6) In some embodiments of any of A1-A5, populating the merge candidate list comprises inserting the reference vector in the merge candidate list before or after a second reference vector corresponding to the merge candidate. For example, refined merge candidate is inserted right before or right after the merge candidate generated without the refinement procedure. In some embodiments, the indicator indicates that the reference vector is to be inserted in the merge candidate list, and the relative position of the reference vector is predefined (e.g., predefined as being before, or after, the second reference vector). In some embodiments, the indicator indicates whether the reference vector is to be inserted before or after the second reference vector. In some embodiments, the reference vector replaces the second reference vector in the merge candidate list. In some embodiments, populating the merge candidate list

comprises inserting the reference vector before one or more other candidates in the merge candidate list.

[0127] (A7) In some embodiments of any of A1-A6, the reference vector is inserted into the merge candidate list when a template-matching cost for the template-matching technique meets one or more criteria. For example, a threshold value may be applied to determine whether the refined merge candidate is constructed in the merge list or not. The refined merge candidate can be constructed in the merge list if the template-matching cost is smaller than and/or equal to the threshold value. In some embodiments, the reference vector is inserted into the merge candidate list in accordance with a determination that the template-matching cost for the template-matching technique meets the one or more criteria. For example, the template-matching cost may be an SAD, SAT), and/or SSE.

[0128] (A8) In some embodiments of A7, the one or more criteria are predefined. For example, the threshold value can be a predefined value.

[0129] (A9) In some embodiments of A8, the one or criteria are selected from a set of predefined criteria based on at least one of a quantization parameter, and a block size. For example, the threshold value can be a quantization parameter (QP) dependent predefined value, meaning that this threshold value can be changed according to the block QP. In some embodiments, the QP is a QP of the current block. In some embodiments, the QP is a QP of the merge candidate or refined merge candidate. In some embodiments, the one or criteria are selected from a set of predefined criteria based on a block size. For example, the threshold value can be a block size dependent value. In some embodiments, the block size corresponds to the current block, the merge candidate, and/or the refined merge candidate.

[0130] (A10) In some embodiments of A7 or A8, the one or more criteria are based on coding information associated with the video bitstream. For example, the threshold value can be a coding mode dependent value. As an example, the one or more criteria are based on a coding mode, a block size, a quantization parameter, and/or other coding information.

[0131] (A11) In some embodiments of any of A7-A10, a second reference vector for the merge candidate is inserted into the merge candidate list when the template-matching cost for the template-matching technique does not meet the one or more criteria. For example, an adaptive selection between refined merge candidate and non-refined merge candidate (which is derived by using original method) may be applied to adaptively select one of the above two candidates to be used to construct the merge candidate. As an example, the refined merge candidate is selected to construct the merge list when the template-matching cost is smaller than and/or equal to the threshold value; otherwise, the non-refined merge candidate is selected for the merge list construction.

[0132] (A12) In some embodiments of any of A1-A11, the template-matching technique is applied within a search window of the merge candidate, and wherein the reference vector points to a reference block in a different picture than the current block. For example, the motion vector may be derived based on the motion information derivation on the refined merge candidate. An example is shown in FIG. 4B, in which the merge candidate A is derived by using the merge list construction procedure, a refined merge candidate is derived using template-matching, and the derived motion

vector of the refined merge candidate is equal to MVA'. In some embodiments, the search window comprises a search area within a current picture. In some embodiments, the search area corresponds to a reconstructed area of the current picture. In some embodiments, the search area has a fixed size. For example, a search range restriction can be applied to the intra template-matching within a search range with fixed size, such as within a current CTU row, or within the current CTU row and N previous coded CTU row(s).

[0133] (A13) In some embodiments of A12, the reference vector is derived from a motion vector of the refined merge candidate and a displacement vector. For example, the motion vector may be derived based on the motion information derivation on the refined merge candidate. An example is shown in FIG. 4C, in which the derived motion vector at the refined merge candidate is equal to MVA'+ BV+DV during the merge list construction. In some embodiments, a displacement vector is applied between the current block and the merge candidate. For example, the final merge motion vector may be equal to the motion vector information derived from the merge candidate and the displacement vector.

[0134] (A14) In some embodiments of A12, the reference vector is derived from a motion vector of the refined merge candidate and a block vector corresponding to the template-matching technique. In some embodiments, the method includes parsing an indicator in the video bitstream, the indicator indicating whether the reference vector is derived using the block vector corresponding to the template-matching technique. For example, a flag is signaled in high-level syntax to indicate whether the block vector is used for lookahead motion vector predictor construction.

[0135] (A15) In some embodiments of any of A1-A14, the reference vector is identified by checking at least one position of the refined merge candidate. For example, the derived motion vector MVA' may be derived from a predefined motion field within the W×H block A' (e.g., as illustrated in FIG. 4D). The availability of motion vector within the motion field may be checked by scanning the availability of motion vector at not less than one position and this scanning order may be a predefined order if multiple positions are checked.

[0136] (A16) In some embodiments of any of A1-A15, the method further comprises identifying a template-matching type from a set of template-matching types, the template-matching technique is applied using the template-matching type. For example, different template-matching types can be used for the intra template-matching process. As an example, the set of template-matching types may include an only-left template, an only-top template, and/or a top-left template. As an example, for smaller blocks (e.g., smaller than or equal to 64 samples), a template size of 2 lines may be used, while for larger blocks a template size of 4 lines may be used. In some embodiments, the different template-matching types are used adaptively, e.g., at a block level. In some embodiments, a syntax is signaled to indicate which template type is used.

[0137] (A17) In some embodiments of any of A1-A16, the template-matching technique uses a subsampled template. For example, pixel subsampling within the template can be used during the template-matching process to calculate the template-matching cost. In some embodiments, the template

matching process is performed at a coarse step size. For example, the step of search is changed to two samples per iteration instead of one.

[0138] (A18) In some embodiments of any of A1-A17, the reference vector is a block vector; and the method further comprises identifying a reference block using the block vector; and identifying a second reference vector for the reference block, the merge candidate list is populated with the second reference vector. For example, a chained motion vector which includes block vector (BV) and/or motion vector can be used to derive the merge candidate by using intra template-matching, as illustrated in FIG. 4E and FIG. 4F. In some embodiments, the second reference block is identified in accordance with an allowed chain depth. For example, the depth of the chained motion vector propagation is limited (e.g., restricted to 1, 2, or 3 block vectors). In some embodiments, adding the block vector it is not considered to increase the depth of the chained motion vector propagation. In some embodiments, subsequent reference blocks are identified in accordance with corresponding block vectors until a chain depth limit reached. In some embodiments, in accordance with the chain depth limit being reached, the merge candidate list is not populated using the reference vector chain. In some embodiments, the combined vector is clipped to be within a predefined area. For example, the chained MV is applied with a clipping operation to ensure that the MV is within a predefined restricted area in the reference picture. In some embodiments, a refinement is applied to the combined vector prior to insertion into the merge candidate list. In some embodiments, a refinement amount is signaled in the video bitstream. In some embodiments, the refinement amount is predefined for the current block.

[0139] (B1) In another aspect, some embodiments include a method (e.g., the method **650**) of video encoding. In some embodiments, the method is performed at a computing system (e.g., the server system **112**) having memory and control circuitry. In some embodiments, the method is performed at a coding module (e.g., the coding module **320**). The method includes (i) receiving video data (e.g., a source video sequence) comprising a current picture composed of a plurality of blocks, including a current block, (ii) identifying a merge candidate for the current block, (iii) identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique, (iv) identifying a reference vector for the refined merge candidate, (v) populating a merge candidate list for the current block using the reference vector, and (vi) encoding the current block using information from the merge candidate list. Some embodiments of B1 include applying any of the techniques described above in A2-A18.

[0140] (C1) In another aspect, some embodiments include a method of visual media data processing. In some embodiments, the method is performed at a computing system (e.g., the server system **112**) having memory and control circuitry. In some embodiments, the method is performed at a coding module (e.g., the coding module **320**). The method includes: (i) receiving video data comprising a current picture composed of a plurality of blocks, including a current block, (ii) identifying a merge candidate for the current block, (iii) identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique, (iv) identifying a reference vector for the refined merge candidate, (v) populating a merge candidate list for

the current block using the reference vector, and (vi) encoding the current block using information from the merge candidate list.

[0141] (D1) In another aspect, some embodiments include a method of processing visual media data. In some embodiments, the method is performed at a computing system (e.g., the server system **112**) having memory and control circuitry. In some embodiments, the method is performed at a coding module (e.g., the coding module **320**). In some embodiments, the method is performed at a source coding component (e.g., the source coder **202**), a coding engine (e.g., the coding engine **212**), and/or an entropy coder (e.g., the entropy coder **214**). The method includes (i) obtaining a source video sequence that comprises a plurality of frames; and (ii) performing a conversion between the source video sequence and a video bitstream of visual media data according to a format rule, where the video bitstream comprises a plurality of encoded blocks including a current block; and where the format rule specifies that: (a) a merge candidate is to be identified for the current block; (b) a refined merge candidate is to be identified by refining a position of the merge candidate using a template-matching technique; (c) a reference vector is to be identified for the refined merge candidate; (d) a merge candidate list is to be populated for the current block using the reference vector; and (e) the current block is to be reconstructed using information from the merge candidate list.

[0142] (E1) In another aspect, some embodiments include a method of video decoding. In some embodiments, the method is performed at a computing system (e.g., the server system **112**) having memory and control circuitry. In some embodiments, the method is performed at a coding module (e.g., the coding module **320**). The method includes (i) receiving a video bitstream comprising a current picture composed of a plurality of blocks, including a current block, (ii) identifying a merge candidate for the current block, (iii) identifying a reference vector for the merge candidate; (iv) generating a combined reference vector by combining the reference vector with a displacement vector, (v) populating a merge candidate list for the current block using the combined reference vector, and (vi) reconstructing the current block using information from the merge candidate list. For example, a displacement vector is applied between current block and the merge candidate to construct the merged motion vector with this displacement information. More specifically, the final merge motion vector is equal to the motion vector information derived from the merge candidate and the said displacement vector.

[0143] (E2) In some embodiments of E1, the merge candidate is a spatial merge candidate. For example, the merge candidate can be any kind of spatial merge candidate.

[0144] (E3) In some embodiments of E1 or E2, the displacement vector is derived based on a displacement offset between a position of the current block and a position of the merge candidate. For example, the displacement vector may be derived based on the displacement offset between the current block position and the merge candidate position.

[0145] In another aspect, some embodiments include a computing system (e.g., the server system **112**) including control circuitry (e.g., the control circuitry **302**) and memory (e.g., the memory **314**) coupled to the control circuitry, the memory storing one or more sets of instructions configured to be executed by the control circuitry, the one or more sets

of instructions including instructions for performing any of the methods described herein (e.g., A1-A18, B1, C1, D1, and E1-E3 above).

[0146] In yet another aspect, some embodiments include a non-transitory computer-readable storage medium storing one or more sets of instructions for execution by control circuitry of a computing system, the one or more sets of instructions including instructions for performing any of the methods described herein (e.g., A1-A18, B1, C1, D1, and E1-E3 above). In some embodiments, a non-transitory computer-readable storage medium stores a video bitstream that is generated by any of the video encoding methods described herein.

[0147] Unless otherwise specified, any of the syntax elements (e.g., indicators) described herein may be high-level syntax (HLS). As used herein, HLS is signaled at a level that is higher than a block level. For example, HLS may correspond to a sequence level, a frame level, a slice level, or a tile level. As another example, HLS elements may be signaled in a video parameter set (VPS), a sequence parameter set (SPS), a picture parameter set (PPS), an adaptation parameter set (APS), a slice header, a picture header, a tile header, and/or a CTU header.

[0148] It will be understood that, although the terms “first,” “second,” etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the claims. As used in the description of the embodiments and the appended claims, the singular forms “a,” “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0149] As used herein, the term “if” can be construed to mean “when” or “upon” or “in response to determining” or “in accordance with a determination” or “in response to detecting” that a stated condition precedent is true, depending on the context. Similarly, the phrase “if it is determined [that a stated condition precedent is true]” or “if [a stated condition precedent is true]” or “when [a stated condition precedent is true]” can be construed to mean “upon determining” or “in response to determining” or “in accordance with a determination” or “upon detecting” or “in response to detecting” that the stated condition precedent is true, depending on the context.

[0150] The foregoing description, for purposes of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be exhaustive or limit the claims to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain principles of operation and practical applications, to thereby enable others skilled in the art.

What is claimed is:

1. A method of video decoding performed at a computing system having memory and one or more processors, the method comprising:

receiving a video bitstream comprising a current picture composed of a plurality of blocks, including a current block;

identifying a merge candidate for the current block;

identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique;

identifying a reference vector for the refined merge candidate;

populating a merge candidate list for the current block using the reference vector; and

reconstructing the current block using information from the merge candidate list.

2. The method of claim 1, wherein the refined merge candidate comprises an adjacent or non-adjacent spatial neighboring block to the current block.

3. The method of claim 1, wherein the refined merge candidate comprises a block co-located with the current block in a reference picture.

4. The method of claim 1, further comprising parsing an indicator in the video bitstream, the indicator indicating whether merge candidate refinement is enabled for the current block, wherein the refined merge candidate is identified when the indicator indicates that merge candidate refinement is enabled for the current block.

5. The method of claim 1, further comprising parsing an indicator in the video bitstream, the indicator indicating whether the refined merge candidate is to be included in the merge candidate list.

6. The method of claim 1, wherein populating the merge candidate list comprises inserting the reference vector in the merge candidate list before or after a second reference vector corresponding to the merge candidate.

7. The method of claim 1, wherein the reference vector is inserted into the merge candidate list when a template-matching cost for the template-matching technique meets one or more criteria.

8. The method of claim 7, wherein the one or more criteria are predefined.

9. The method of claim 8, wherein the one or more criteria are selected from a set of predefined criteria based on at least one of a quantization parameter, and a block size.

10. The method of claim 7, wherein the one or more criteria are based on coding information associated with the video bitstream.

11. The method of claim 7, wherein a second reference vector for the merge candidate is inserted into the merge candidate list when the template-matching cost for the template-matching technique does not meet the one or more criteria.

12. The method of claim 1, wherein the template-matching technique is applied within a search window of the merge candidate, and wherein the reference vector points to a reference block in a different picture than the current block.

13. The method of claim 12, wherein the reference vector is derived from a motion vector of the refined merge candidate and a displacement vector.

14. The method of claim 12, wherein the reference vector is derived from a motion vector of the refined merge candidate and a block vector corresponding to the template-matching technique.

15. The method of claim 1, wherein the reference vector is identified by checking at least one position of the refined merge candidate.

16. The method of claim 1, further comprising identifying a template-matching type from a set of template-matching types, wherein the template-matching technique is applied using the template-matching type.

17. The method of claim 1, wherein the template-matching technique uses a subsampled template.

18. The method of claim 1, wherein the reference vector is a block vector; and

wherein the method further comprises:

identifying a reference block using the block vector; and

identifying a second reference vector for the reference block, wherein the merge candidate list is populated with the second reference vector.

19. A method of video encoding performed at a computing system having memory and one or more processors, the method comprising:

receiving video data comprising a current picture composed of a plurality of blocks, including a current block;

identifying a merge candidate for the current block;

identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique;

identifying a reference vector for the refined merge candidate;

populating a merge candidate list for the current block using the reference vector; and

encoding the current block using information from the merge candidate list.

20. A non-transitory computer-readable storage medium storing a video bitstream that is generated by a video encoding method, the video encoding method comprising:

receiving video data comprising a current picture composed of a plurality of blocks, including a current block;

identifying a merge candidate for the current block;

identifying a refined merge candidate by refining a position of the merge candidate using a template-matching technique;

identifying a reference vector for the refined merge candidate;

populating a merge candidate list for the current block using the reference vector; and

encoding the current block using information from the merge candidate list.

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